

6 Inner Product Spaces

6.1 Inner Products and Norms

You should be familiar with the *scalar* or *dot product* from previous courses, and with its basic properties: for instance, in $\mathbb{R}^2$,

$$\begin{pmatrix} x \\ y \end{pmatrix} \cdot \begin{pmatrix} z \\ w \end{pmatrix} = xz + yw$$

Moreover, the *angle* θ between two vectors satisfies the formula

$$\mathbf{x} \cdot \mathbf{y} = \|\mathbf{x}\| \|\mathbf{y}\| \cos \theta$$

where $\|\mathbf{x}\| = \sqrt{\mathbf{x} \cdot \mathbf{x}}$ is the *norm* or length of the vector $\mathbf{x}$. The dot product is therefore associated to the idea of *orthogonality* or *perpendicularity*:

$$\mathbf{x} \cdot \mathbf{y} = 0 \iff \mathbf{x}, \mathbf{y} \text{ are perpendicular}$$

The dot product is precisely the algebraic structure that allows us to consider length and angle. An *inner product* is a generalization that does exactly the same thing for a vector space.

Definition 6.1. An *inner product space* is a vector space V over a field $\mathbb{F}$ ($= \mathbb{R}$ or $\mathbb{C}$) equipped with an *inner product*, a function $\langle \cdot, \cdot \rangle : V \times V \rightarrow \mathbb{F}$ satisfying the following properties: $\forall \lambda \in \mathbb{F}, \mathbf{x}, \mathbf{y}, \mathbf{z} \in V$,

$$(a) \langle \lambda \mathbf{x} + \mathbf{y}, \mathbf{z} \rangle = \lambda \langle \mathbf{x}, \mathbf{z} \rangle + \langle \mathbf{y}, \mathbf{z} \rangle$$

$$(b) \overline{\langle \mathbf{x}, \mathbf{y} \rangle} = \langle \mathbf{y}, \mathbf{x} \rangle$$

$$(c) \mathbf{x} \neq \mathbf{0} \implies \langle \mathbf{x}, \mathbf{x} \rangle > 0$$

The *norm* or *length* of a vector $\mathbf{x}$ is defined to be $\|\mathbf{x}\| = \sqrt{\langle \mathbf{x}, \mathbf{x} \rangle}$: a *unit vector* has $\|\mathbf{x}\| = 1$.

Vectors $\mathbf{x}, \mathbf{y}$ are *perpendicular* or *orthogonal* if $\langle \mathbf{x}, \mathbf{y} \rangle = 0$. Vectors are *orthonormal* if they are orthogonal unit vectors.

Property (a) says that an inner product is linear in its first entry: for any fixed $\mathbf{z} \in V$, the map

$$L_{\mathbf{z}} : V \rightarrow \mathbb{F} : \mathbf{x} \mapsto \langle \mathbf{x}, \mathbf{z} \rangle$$

is linear. The bar in property (b) is the complex conjugate (only necessary if $\mathbb{F} = \mathbb{C}$). This allows us to consider linear combinations in the second entry:

$$\langle \mathbf{x}, \lambda \mathbf{y} + \mathbf{z} \rangle = \overline{\langle \lambda \mathbf{y} + \mathbf{z}, \mathbf{x} \rangle} = \overline{\lambda \langle \mathbf{y}, \mathbf{x} \rangle + \langle \mathbf{z}, \mathbf{x} \rangle} = \overline{\lambda} \overline{\langle \mathbf{y}, \mathbf{x} \rangle} + \overline{\langle \mathbf{z}, \mathbf{x} \rangle} = \overline{\lambda} \langle \mathbf{x}, \mathbf{y} \rangle + \langle \mathbf{x}, \mathbf{z} \rangle \quad (*)$$

This ‘conjugate-linear’ property is sometimes known as *sesquilinearity*: the prefix *sesqui* meaning ‘one and a half.’ If $\mathbb{F} = \mathbb{R}$, then the inner product is linear in both slots and is termed *bilinear*.

Finally, property (c) is known as *positive definiteness*: this forces the norm of a non-zero vector to be positive.

Warning! In case you dabble in the dark arts of Physics, be aware that its convention is for an inner product to be conjugate-linear in the *first* entry rather than the second!

Examples 6.2. 1. In $\mathbb{F}^n$, the *standard inner product* is^a

$$\langle \mathbf{x}, \mathbf{y} \rangle = \mathbf{y}^* \mathbf{x} = \sum_{j=1}^n x_j \overline{y_j} = x_1 \overline{y_1} + \cdots + x_n \overline{y_n}$$

Properties (a)–(c) should be obvious. When $\mathbb{F} = \mathbb{C}$ this is known as the standard *Hermitian product*, while in $\mathbb{R}^n$ it is the *dot product*. The *norm* is the well-known length function:^b

$$\|\mathbf{x}\| = \sqrt{\sum_{j=1}^n |x_j|^2}$$

2. We modify the previous example with *weights*: let $a_1, \dots, a_n$ be positive real numbers and define

$$\langle \mathbf{x}, \mathbf{y} \rangle = \sum_{j=1}^n a_j x_j \overline{y_j} = a_1 x_1 \overline{y_1} + \cdots + a_n x_n \overline{y_n}$$

Check the inner product properties:

- (a) $\langle \lambda \mathbf{x} + \mathbf{y}, \mathbf{z} \rangle = \sum_{j=1}^n a_j (\lambda x_j + y_j) \overline{z_j} = \lambda \sum_{j=1}^n a_j x_j \overline{z_j} + \sum_{j=1}^n a_j y_j \overline{z_j} = \lambda \langle \mathbf{x}, \mathbf{z} \rangle + \langle \mathbf{y}, \mathbf{z} \rangle$
- (b) $\overline{\langle \mathbf{x}, \mathbf{y} \rangle} = \sum_{j=1}^n a_j \overline{x_j} \overline{\overline{y_j}} = \langle \mathbf{y}, \mathbf{x} \rangle$: we used the fact that $a_j \in \mathbb{R}$ here.
- (c) $\langle \mathbf{x}, \mathbf{x} \rangle = \sum_{j=1}^n a_j |x_j|^2 > 0$ whenever $\mathbf{x} \neq \mathbf{0}$: this is the reason we need *all* $a_j > 0$.

Note that different inner product create different relationships between vectors. For instance, compare the standard ‘dot’ product on $\mathbb{R}^2$ with the alternative inner product

$$\langle \mathbf{x}, \mathbf{y} \rangle = x_1 y_1 + 3x_2 y_2$$

With respect to these distinct inner products, vectors have different properties: for example if $\mathbf{x} = \frac{1}{2} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ and $\mathbf{y} = \frac{1}{2\sqrt{3}} \begin{pmatrix} 3 \\ -1 \end{pmatrix}$, we see that $\{\mathbf{x}, \mathbf{y}\}$ is an orthonormal set with respect to $\langle, \rangle$:

$$\begin{aligned} \|\mathbf{x}\|^2 &= \frac{1}{4}(1^2 + 3 \cdot 1^2) = 1, & \|\mathbf{y}\|^2 &= \frac{1}{12}(3^2 + 3 \cdot 1^2) = 1, \\ \langle \mathbf{x}, \mathbf{y} \rangle &= \frac{1}{4\sqrt{3}}(3 - 3) = 0 \end{aligned}$$

With the dot product, these vectors are not at all special:

$$\mathbf{x} \cdot \mathbf{x} = \frac{1}{2}, \quad \mathbf{y} \cdot \mathbf{y} = \frac{5}{6}, \quad \mathbf{x} \cdot \mathbf{y} = \frac{1}{2\sqrt{3}}$$

We have the same *vector space* $\mathbb{R}^2$, but different *inner product spaces*: $(\mathbb{R}^2, \cdot) \neq (\mathbb{R}^2, \langle, \rangle)$.

^a $\mathbf{y}^* = \overline{\mathbf{y}}^T$ is the conjugate transpose of $\mathbf{y}$.

^bWhile unnecessary when $\mathbb{F} = \mathbb{R}$, the expression $|x_j| = \sqrt{x_j \overline{x_j}}$ denotes the *modulus* of a complex number when $\mathbb{F} = \mathbb{C}$. It is to distinguish between this and the norm that we use slightly different notation for each!

We’ve already seen one of the basic properties of an inner product; we restate here for reference.

Lemma 6.3. Let $\langle \cdot, \cdot \rangle$ be an inner product on V over $\mathbb{F}$ with norm $\|\cdot\|$. Let $\mathbf{x}, \mathbf{y}, \mathbf{z} \in V$ and $\lambda \in \mathbb{F}$.

1. (Conjugate-linearity (*)) $\langle \mathbf{x}, \lambda \mathbf{y} + \mathbf{z} \rangle = \overline{\lambda} \langle \mathbf{x}, \mathbf{y} \rangle + \langle \mathbf{x}, \mathbf{z} \rangle$
2. $\langle \mathbf{x}, \mathbf{0} \rangle = 0$
3. $\|\mathbf{x}\| = 0 \iff \mathbf{x} = \mathbf{0}$
4. $\|\lambda \mathbf{x}\| = |\lambda| \|\mathbf{x}\|$
5. $\langle \mathbf{x}, \mathbf{y} \rangle = \langle \mathbf{x}, \mathbf{z} \rangle$ for all $\mathbf{x} \implies \mathbf{y} = \mathbf{z}$
6. (Cauchy-Schwarz inequality) $|\langle \mathbf{x}, \mathbf{y} \rangle| \leq \|\mathbf{x}\| \|\mathbf{y}\|$ with equality if and only if $\mathbf{x}$ and $\mathbf{y}$ are parallel
7. (Triangle Inequality) $\|\mathbf{x} + \mathbf{y}\| \leq \|\mathbf{x}\| + \|\mathbf{y}\|$ with equality if and only if $\mathbf{x}$ and $\mathbf{y}$ are parallel and point in the same direction

Proof. Part 1 is done above; we leave 2–4 as exercises.

5. Let $\mathbf{x} = \mathbf{y} - \mathbf{z}$ to see that $\langle \mathbf{x}, \mathbf{y} \rangle = \langle \mathbf{x}, \mathbf{z} \rangle \implies \|\mathbf{y} - \mathbf{z}\|^2 = 0 \implies \mathbf{y} = \mathbf{z}$ by part 2.
6. If $\mathbf{y} = \mathbf{0}$, the result is trivial. WLOG we may assume $\|\mathbf{y}\| = 1$: if the inequality holds for this it holds for all $\mathbf{y}$. Expand out the following:

$$\begin{aligned} 0 &\leq \|\mathbf{x} - \langle \mathbf{x}, \mathbf{y} \rangle \mathbf{y}\|^2 = \|\mathbf{x}\|^2 + |\langle \mathbf{x}, \mathbf{y} \rangle|^2 - \overline{\langle \mathbf{x}, \mathbf{y} \rangle} \langle \mathbf{x}, \mathbf{y} \rangle - \langle \mathbf{x}, \mathbf{y} \rangle \langle \mathbf{y}, \mathbf{x} \rangle \\ &= \|\mathbf{x}\|^2 - |\langle \mathbf{x}, \mathbf{y} \rangle|^2 \implies |\langle \mathbf{x}, \mathbf{y} \rangle| \leq \|\mathbf{x}\| \|\mathbf{y}\| \end{aligned}$$

Equality is if and only if $\mathbf{x} = \langle \mathbf{x}, \mathbf{y} \rangle \mathbf{y}$ which is precisely when $\mathbf{x}, \mathbf{y}$ are linearly dependent.

7. We establish the squared result:

$$\begin{aligned} (\|\mathbf{x}\| + \|\mathbf{y}\|)^2 - \|\mathbf{x} + \mathbf{y}\|^2 &= \|\mathbf{x}\|^2 + 2\|\mathbf{x}\| \|\mathbf{y}\| + \|\mathbf{y}\|^2 - \|\mathbf{x}\|^2 - \langle \mathbf{x}, \mathbf{y} \rangle - \langle \mathbf{y}, \mathbf{x} \rangle - \|\mathbf{y}\|^2 \\ &= 2\|\mathbf{x}\| \|\mathbf{y}\| - \langle \mathbf{x}, \mathbf{y} \rangle - \overline{\langle \mathbf{x}, \mathbf{y} \rangle} \\ &= 2\|\mathbf{x}\| \|\mathbf{y}\| - 2\Re \langle \mathbf{x}, \mathbf{y} \rangle && \text{(real part)} \\ &\geq 2(\|\mathbf{x}\| \|\mathbf{y}\| - |\langle \mathbf{x}, \mathbf{y} \rangle|) && (\Re z \leq |z| \text{ for any } z \in \mathbb{C}) \\ &\geq 0 && \text{(Cauchy-Schwarz)} \end{aligned}$$

Equality requires both equality in Cauchy-Schwarz (parallel) and that $\Re \langle \mathbf{x}, \mathbf{y} \rangle \geq 0$: since $\mathbf{x}, \mathbf{y}$ are already parallel, this means that one is a non-negative multiple of the other. ■

The use of complex numbers in some of these proofs/examples may seem difficult. This is quite deliberate but also very easy to get around. It is perfectly acceptable (and encouraged!) to restrict to *real* inner product spaces until you are comfortable: try rewriting the above proofs in this context and see how much easier they are!

Further Examples of Inner Products

The Frobenius inner product For any $A, B \in M_{m \times n}(\mathbb{F})$, define

$$\langle A, B \rangle = \text{tr}(B^* A)$$

where tr is the *trace* of a square matrix, and $B^* = \overline{B}^T$ is the *conjugate transpose*. This makes $M_{m \times n}(\mathbb{F})$ into an inner product space. When $n = 1$, this is simply the standard Hermitian inner product on $\mathbb{C}^n$.

Example 6.4. In $M_{3 \times 2}(\mathbb{C})$,

$$\begin{aligned} \left\langle \begin{pmatrix} 1 & i \\ 2-i & 0 \\ 0 & -1 \end{pmatrix}, \begin{pmatrix} 0 & 7 \\ 1 & 2i \\ 3-2i & 4 \end{pmatrix} \right\rangle &= \text{tr} \begin{pmatrix} 0 & 1 & 3+2i \\ 7 & -2i & 4 \end{pmatrix} \begin{pmatrix} 1 & i \\ 2-i & 0 \\ 0 & -1 \end{pmatrix} = \text{tr} \begin{pmatrix} 2-i & -3-2i \\ 9-4i & -4+7i \end{pmatrix} \\ &= 2-i-4+7i = -2-8i \\ \left\| \begin{pmatrix} 1 & i \\ 2-i & 0 \\ 0 & -1 \end{pmatrix} \right\|^2 &= \text{tr} \begin{pmatrix} 1 & 2+i & 0 \\ -i & 0 & -1 \end{pmatrix} \begin{pmatrix} 1 & i \\ 2-i & 0 \\ 0 & -1 \end{pmatrix} = \text{tr} \begin{pmatrix} 6 & i \\ -i & 2 \end{pmatrix} = 8 \end{aligned}$$

As before, if this example seems too difficult because you are unfamiliar with the complex numbers, first restrict to the reals: if $\mathbb{F} = \mathbb{R}$, then $\langle A, B \rangle = \text{tr}(B^T A)$.

The L^2 inner product If $f, g \in C([a, b])$ are continuous functions on a bounded interval $[a, b]$, define

$$\langle f, g \rangle := \int_a^b f(x) \overline{g(x)} \, dx$$

If you've studied analysis, this should feel legitimate: continuous functions on closed bounded intervals are integrable. The properties of an inner product are easy to check, in particular,

$$\|f\|^2 = \int_a^b |f(x)|^2 \, dx = 0 \iff f(x) \equiv 0$$

follows because f is continuous. With sufficient care, we can even take the interval to be $(-\infty, \infty)$ though we must restrict to functions f for which $\int_{\mathbb{R}} |f(x)| \, dx < \infty$. This is our first example of an *infinite dimensional* inner product space.

Example 6.5. Let $f, g \in C[-1, 1]$ where $f(x) = x$ and $g(x) = x^2$. We compute the L^2 inner products:

$$\langle f, g \rangle = \int_{-1}^1 x^3 \, dx = 0, \quad \|f\|^2 = \int_{-1}^1 x^2 \, dx = \frac{2}{3}, \quad \|g\|^2 = \int_{-1}^1 x^4 \, dx = \frac{2}{5}$$

With some simple scaling, we see that $\left\{ \frac{1}{\|f\|} f, \frac{1}{\|g\|} g \right\} = \left\{ \sqrt{\frac{3}{2}} x, \sqrt{\frac{5}{2}} x^2 \right\}$ is an orthonormal set.

There are many, many more examples in this vein; indeed since every polynomial is continuous, these inner products also turn any vector space of polynomials into an inner product space.

The ℓ^2 inner product (for Analysis aficionados only!) The standard inner product on $\mathbb{F}^n$ can be generalized to the vector space ℓ^2 of *square-summable sequences* with values in $\mathbb{F}$. That is,

$$(s_n) \in \ell^2 \iff \sum_{n=1}^{\infty} |s_n|^2 < \infty$$

The inner product is defined by

$$\langle (s_n), (t_n) \rangle = \sum_{n=1}^{\infty} s_n \overline{t_n}$$

It is a little work, and outside the scope of this course, to check the convergence conditions necessary to prove that ℓ^2 really is a vector space and that we have an inner product. You should compare the structure with the L^2 inner product: this is also an infinite dimensional inner product space. Both inner products can also be generalized with weights analogously to Example 6.1.2.

Exercises. 6.1.1. Define $\langle \mathbf{x}, \mathbf{y} \rangle := \sum_{j=1}^n x_j y_j$ on $\mathbb{C}^n$. Is this an inner product? Which of the properties

(a), (b), (c) of Definition 6.1 does it satisfy?

6.1.2. Prove parts 2, 3 and 4 of Lemma 6.3.

6.1.3. Let V be an inner product space, prove Pythagoras' Theorem:

$$\text{If } \langle \mathbf{x}, \mathbf{y} \rangle = 0 \text{ then } \|\mathbf{x} + \mathbf{y}\|^2 = \|\mathbf{x}\|^2 + \|\mathbf{y}\|^2$$

6.1.4. Let $m \in \mathbb{Z}$ and consider the complex-valued^a function $f_m(x) = \frac{1}{\sqrt{2\pi}} e^{imx}$ defined on the interval $[-\pi, \pi]$. If $\langle \cdot, \cdot \rangle$ is the L^2 inner product on $C([-\pi, \pi])$, prove that $\{f_m : m \in \mathbb{Z}\}$ is an orthonormal set. This example is central to the study of *Fourier series*.

6.1.5. Let $T \in \mathcal{L}(V)$ be a linear operator on an inner product space and suppose that $\|T(\mathbf{x})\| = \|\mathbf{x}\|$ for all $\mathbf{x} \in V$. Prove that T is injective (1-1).

6.1.6. Prove the polarization identities:

$$(a) \text{ In any real inner product space: } \langle \mathbf{x}, \mathbf{y} \rangle = \frac{1}{4} \|\mathbf{x} + \mathbf{y}\|^2 - \frac{1}{4} \|\mathbf{x} - \mathbf{y}\|^2$$

$$(b) \text{ In any complex inner product space: } \langle \mathbf{x}, \mathbf{y} \rangle = \frac{1}{4} \sum_{k=1}^4 i^k \|\mathbf{x} + i^k \mathbf{y}\|^2$$

6.1.7. Let $A, B \in M_n(\mathbb{F})$ and let $\langle \cdot, \cdot \rangle$ be the standard inner product on $\mathbb{F}^n$.

(a) Prove that $\forall \mathbf{x}, \mathbf{y} \in \mathbb{F}^n$, we have $\langle \mathbf{x}, A\mathbf{y} \rangle = \langle A^* \mathbf{x}, \mathbf{y} \rangle$.

(b) Suppose $\langle \mathbf{x}, A\mathbf{y} \rangle = \langle B\mathbf{x}, \mathbf{y} \rangle$ for all $\mathbf{x}, \mathbf{y} \in \mathbb{F}^n$: prove that $B = A^*$.

(c) Let β be any orthonormal basis of $\mathbb{F}^n$ and let Q be the matrix whose columns are the co-ordinates of the elements of β with respect to the standard basis. Prove that $Q^* = Q^{-1}$.

^aIf the reliance on complex numbers is alarming, recall Euler's formula $e^{imx} = \cos mx + i \sin mx$. One can instead work entirely with the real-valued functions $\cos mx$ and $\sin mx$. The problem is that you then need integration by parts :(

6.2 Gram–Schmidt Orthogonalization and Orthogonal Subspaces

This section has two goals: to see how orthogonal sets simplify calculations in an inner product space, and to see how to construct orthogonal sets. Before calculating anything, it is useful to introduce some terminology.

Definition 6.6. Let $\mathbf{w} \in V$ be a non-zero vector in an inner product space V . The *orthogonal projection* onto $\text{Span}\{\mathbf{w}\}$ is the map:

$$\pi_{\mathbf{w}} : V \rightarrow V : \mathbf{x} \mapsto \frac{\langle \mathbf{x}, \mathbf{w} \rangle}{\|\mathbf{w}\|^2} \mathbf{w}$$

More generally, if $U = \text{Span}\{\mathbf{u}_1, \dots, \mathbf{u}_n\}$ is a *finite-dimensional* subspace spanned by non-zero orthogonal vectors $\mathbf{w}_j$, then the orthogonal projection onto U is

$$\pi_U : V \rightarrow V : \mathbf{x} \mapsto \sum_{j=1}^n \pi_{\mathbf{w}_j}(\mathbf{x}) = \sum_{j=1}^n \frac{\langle \mathbf{x}, \mathbf{w}_j \rangle}{\|\mathbf{w}_j\|^2} \mathbf{w}_j$$

Orthogonal projections allow us to easily compute the coefficients of a vector with respect to a basis.

Theorem 6.7. Suppose $S = \{\mathbf{w}_1, \dots, \mathbf{w}_n\}$ is a finite orthogonal subset of V where all $\mathbf{w}_j \neq \mathbf{0}$, let $U = \text{Span } S$, and $\mathbf{y} \in U$. Then

$$\mathbf{y} = \pi_U(\mathbf{y}) = \sum_{j=1}^n \frac{\langle \mathbf{y}, \mathbf{w}_j \rangle}{\|\mathbf{w}_j\|^2} \mathbf{w}_j \quad (*)$$

Moreover, S is a linearly independent set and thus a basis of U .

If S is additionally orthonormal, then the formula is even simpler: $\mathbf{y} = \sum_{j=1}^n \langle \mathbf{y}, \mathbf{w}_j \rangle \mathbf{w}_j$.

Observe also that $\mathbf{y} = \pi_U(\mathbf{y})$ means π_U , at least¹ applied to U , is independent of the choice of orthogonal basis of U .

Proof. If $\mathbf{y} = \sum_{k=1}^n a_k \mathbf{w}_k$, then the linearity of the inner product in the first slot says

$$\langle \mathbf{y}, \mathbf{w}_j \rangle = \sum_{k=1}^n a_k \langle \mathbf{w}_k, \mathbf{w}_j \rangle = a_j \|\mathbf{w}_j\|^2$$

Now apply this to $\mathbf{y} = \mathbf{0}$ to see that the only linear combination of S summing to zero is trivial. ■

It is easy to miss what this theorem is saying: an orthogonal set makes it almost trivial to calculate the coefficients of a linear combination!

¹We shall see later that π_U really is well-defined as a function $\pi_U : V \rightarrow V$. This requires a little more work.

Examples 6.8. 1. Let $S = \{\mathbf{i}, \mathbf{j}\} \leq \mathbb{R}^2$ with the standard dot product, and let $\mathbf{y} = \begin{pmatrix} x \\ y \end{pmatrix}$. The coefficients of the linear combination are clearly

$$\pi_{\mathbf{i}}(\mathbf{y}) = \frac{\langle \mathbf{y}, \mathbf{i} \rangle}{\|\mathbf{i}\|^2} = x, \quad \pi_{\mathbf{j}}(\mathbf{y}) = \frac{\langle \mathbf{y}, \mathbf{j} \rangle}{\|\mathbf{j}\|^2} = y$$

2. Observe that $S = \{\mathbf{w}_1, \mathbf{w}_2, \mathbf{w}_3\} = \left\{ \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}, \begin{pmatrix} 2 \\ -1 \\ 0 \end{pmatrix}, \begin{pmatrix} 3 \\ 6 \\ -5 \end{pmatrix} \right\}$ is an orthogonal set (indeed a basis) in $\mathbb{R}^3$ with respect to the standard (dot) inner product. Now let $\mathbf{y} = \begin{pmatrix} 7 \\ 4 \\ 2 \end{pmatrix}$ and compute:

$$\begin{aligned} \mathbf{y} &= \sum_{j=1}^3 \frac{\langle \mathbf{y}, \mathbf{w}_j \rangle}{\|\mathbf{w}_j\|^2} \mathbf{w}_j = \frac{7+8+6}{1+4+9} \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} + \frac{14-4+0}{4+1+0} \begin{pmatrix} 2 \\ -1 \\ 0 \end{pmatrix} + \frac{21+24-10}{9+36+25} \begin{pmatrix} 3 \\ 6 \\ -5 \end{pmatrix} \\ &= \frac{3}{2} \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} + 2 \begin{pmatrix} 2 \\ -1 \\ 0 \end{pmatrix} + \frac{1}{2} \begin{pmatrix} 3 \\ 6 \\ -5 \end{pmatrix} \implies [\mathbf{y}]_S = \begin{pmatrix} 3/2 \\ 2 \\ 1/2 \end{pmatrix} \end{aligned}$$

Compare this with the slow (augmented matrix) method for finding the coefficients!

The Gram–Schmidt process

Since orthogonal sets are so useful for simplifying calculations, the burning question is then *how to find them?* We now describe an algorithm for constructing an orthogonal set from a given ordered set in such a way that the spans of the two sets are the same. The trick is to employ repeated projections.

Theorem 6.9 (Gram–Schmidt). Suppose $S = \{\mathbf{s}_1, \dots, \mathbf{s}_n\}$ is a linearly independent subset of an inner product space V . Construct a sequence of vectors $\mathbf{t}_1, \dots, \mathbf{t}_n$ inductively:

- Define $\mathbf{t}_1 = \mathbf{s}_1$
- For each $k \geq 2$, define $\mathbf{t}_k = \mathbf{s}_k - \pi_{\text{Span}\{\mathbf{t}_1, \dots, \mathbf{t}_{k-1}\}}(\mathbf{s}_k) = \mathbf{s}_k - \sum_{j=1}^{k-1} \frac{\langle \mathbf{s}_k, \mathbf{t}_j \rangle}{\|\mathbf{t}_j\|^2} \mathbf{t}_j$ (*)

Then $T := \{\mathbf{t}_1, \dots, \mathbf{t}_n\}$ is an orthogonal set with $\text{Span } T = \text{Span } S$.

Example 6.10. Let $S = \{\mathbf{s}_1, \mathbf{s}_2, \mathbf{s}_3\} = \left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \right\} \subseteq \mathbb{R}^3$. We compute:

$$\mathbf{t}_1 = \mathbf{s}_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \implies \|\mathbf{t}_1\|^2 = 1$$

$$\mathbf{t}_2 = \mathbf{s}_2 - \pi_{\mathbf{t}_1}(\mathbf{s}_2) = \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix} - \frac{2}{1} \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ -1 \\ 3 \end{pmatrix} \implies \|\mathbf{t}_2\|^2 = 10$$

$$\mathbf{t}_3 = \mathbf{s}_3 - \pi_{\text{Span}\{\mathbf{t}_1, \mathbf{t}_2\}}(\mathbf{s}_3) = \mathbf{s}_3 - \frac{\langle \mathbf{s}_3, \mathbf{t}_1 \rangle}{\|\mathbf{t}_1\|^2} \mathbf{t}_1 - \frac{\langle \mathbf{s}_3, \mathbf{t}_2 \rangle}{\|\mathbf{t}_2\|^2} \mathbf{t}_2 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} - \frac{1}{1} \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} - \frac{2}{10} \begin{pmatrix} 0 \\ -1 \\ 3 \end{pmatrix} = \begin{pmatrix} 0 \\ 6/5 \\ 2/5 \end{pmatrix}$$

The orthogonality of $T = \{\mathbf{t}_1, \mathbf{t}_2, \mathbf{t}_3\}$ should be clear, as should $\text{Span}\{\mathbf{t}_1, \mathbf{t}_2\} = \text{Span}\{\mathbf{s}_1, \mathbf{s}_2\}$. One could easily produce a simpler orthogonal basis of $\mathbb{R}^3$ by scaling, or indeed an orthonormal basis.

Proof of Theorem 6.9. We prove by induction. For each $i \leq n$, define $S_i = \{\mathbf{s}_1, \dots, \mathbf{s}_i\}$ and $T_i = \{\mathbf{t}_1, \dots, \mathbf{t}_i\}$.

Certainly $T_1 = \{\mathbf{t}_1\} = S_1$ is an orthogonal set.

Fix $k \geq 2$, assume T_{k-1} is orthogonal and that $\text{Span } T_{k-1} = \text{Span } S_{k-1}$. If $\mathbf{t}_k = \mathbf{0}$, (*) says that

$$\mathbf{s}_k \in \text{Span } T_{k-1} = \text{Span } S_{k-1}$$

and so S_k is not linearly independent: contradiction. We conclude that $\mathbf{t}_k \neq \mathbf{0}$.

Now suppose $i < k$: by the linearity of the inner product, we have

$$\langle \mathbf{t}_k, \mathbf{t}_i \rangle = \langle \mathbf{s}_k, \mathbf{t}_i \rangle - \sum_{j=1}^{k-1} \frac{\langle \mathbf{s}_k, \mathbf{t}_j \rangle}{\|\mathbf{t}_j\|^2} \langle \mathbf{t}_j, \mathbf{t}_i \rangle = \langle \mathbf{s}_k, \mathbf{t}_i \rangle - \frac{\langle \mathbf{s}_k, \mathbf{t}_i \rangle}{\|\mathbf{t}_i\|^2} \langle \mathbf{t}_i, \mathbf{t}_i \rangle = 0$$

where the orthogonality of T_{k-1} was used to see that most of the inner products vanish. We see that $\mathbf{t}_k$ is a non-zero vector orthogonal to T_{k-1} , whence T_k is an orthogonal set.

Moreover, $\mathbf{s}_k \in \text{Span } T_k \implies \text{Span } S_k \leq \text{Span } T_k$. Similarly, $\mathbf{t}_k \in \text{Span}(\{\mathbf{s}_k\} \cup T_{k-1}) = \text{Span } S_k$, by assumption and we conclude that $\text{Span } T_k \leq \text{Span } S_k$, which completes the theorem. ■

Example 6.11. This time we work in the inner product space of real-valued continuous functions $C([0, 1])$ equipped with the L^2 inner product $\langle f, g \rangle = \int_0^1 f(x)g(x) dx$.

Let $S = \{1, x, x^2\}$ and apply the algorithm:

$$\begin{aligned} t_1 = 1 &\implies \|t_1\|^2 = \int_0^1 1 dx = 1 \\ t_2 = x - \pi_{t_1}(x) &= x - \frac{\langle x, 1 \rangle}{\|1\|^2} 1 = x - \int_0^1 x dx = x - \frac{1}{2} \\ &\implies \|t_2\|^2 = \int_0^1 \left(x - \frac{1}{2}\right)^2 dx = \frac{1}{12} \\ t_3 = x^2 - \pi_{\text{Span}\{t_1, t_2\}}(x^2) &= x^2 - \frac{\langle x^2, 1 \rangle}{\|1\|^2} 1 - \frac{\langle x^2, x - \frac{1}{2} \rangle}{\|x - \frac{1}{2}\|^2} \left(x - \frac{1}{2}\right) \\ &= x^2 - \int_0^1 x^2 dx - \frac{\int_0^1 x^2 (x - \frac{1}{2}) dx}{1/12} \left(x - \frac{1}{2}\right) = x^2 - x + \frac{1}{6} \end{aligned}$$

This example can be extended to arbitrary degree since $\{1, x, x^2, \dots\}$ is an infinite linearly independent family in $C([0, 1])$.

By applying Gram–Schmidt to any basis, we instantly see the following:

Corollary 6.12. Every finite-dimensional inner product space has an orthonormal basis.

Orthogonal Complements and Projections

As an aid to our intuition, it is worth returning to the notion of orthogonal projection (Definition 6.6). It is useful to introduce a little more language to help our intuition.

Definition 6.13. Let U be a subspace of an inner product space V . Its *orthogonal complement* $U^\perp$ (read U -perp) is the set of all vectors orthogonal to U :

$$U^\perp := \{\mathbf{x} \in V : \forall \mathbf{w} \in U, \langle \mathbf{x}, \mathbf{w} \rangle = 0\}$$

The most basic properties of U are an easy exercise.

Lemma 6.14. $U^\perp$ is a subspace of V and $U \cap U^\perp = \{\mathbf{0}\}$.

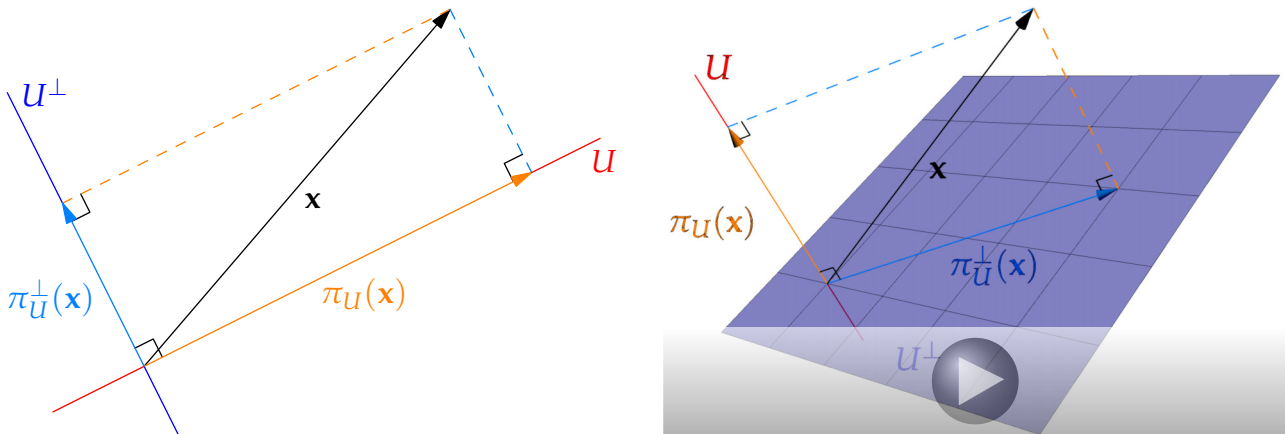
Example 6.15. Following Example 6.10, the subspace $U = \text{Span} \left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 3 \end{pmatrix} \right\}$ of $\mathbb{R}^3$ has orthogonal complement $U^\perp = \text{Span} \left\{ \begin{pmatrix} 0 \\ 3 \\ 1 \end{pmatrix} \right\}$.

Definition 6.16. Suppose U is a finite-dimensional subspace of an inner product space V : by virtue of Corollary 6.12, the function $\pi_U : V \rightarrow V$ may be defined using any orthonormal basis of U . The *perpendicular* or *complementary projection* is the function

$$\pi_U^\perp : V \rightarrow V : \mathbf{x} \mapsto \mathbf{x} - \pi_U(\mathbf{x})$$

We now get to see the origin of the term *orthogonal projection*: in $\mathbb{R}^2$ and $\mathbb{R}^3$, the vectors $\pi_U(\mathbf{x})$ and $\pi_U^\perp(\mathbf{x})$ form the sides of a right triangle with hypotenuse $\mathbf{x}$, so that the two projections form an orthogonal decomposition:

$$\mathbf{x} = \pi_U(\mathbf{x}) + \pi_U^\perp(\mathbf{x}) \quad \text{and} \quad \pi_U(\mathbf{x}) \perp \pi_U^\perp(\mathbf{x})$$



We finally check that everything goes through in a general inner product space.

Theorem 6.17. The functions π_U and $\pi_U^\perp$ are linear maps satisfying

$$\pi_U(\mathbf{x}) = \begin{cases} \mathbf{x} & \text{if } \mathbf{x} \in U \\ \mathbf{0} & \text{if } \mathbf{x} \in U^\perp \end{cases} \quad \pi_U^\perp(\mathbf{x}) = \begin{cases} \mathbf{x} & \text{if } \mathbf{x} \in U^\perp \\ \mathbf{0} & \text{if } \mathbf{x} \in U \end{cases}$$

Moreover $\pi_U^\perp(\mathbf{w}) \in U^\perp$ for any $\mathbf{w} \in V$ and $U \oplus U^\perp = V$.

Finally, $\pi_U(\mathbf{x})$ is the closest vector in the subspace U to $\mathbf{x}$, in that

$$\forall \mathbf{w} \in U, \|\pi_U(\mathbf{x}) - \mathbf{x}\| \leq \|\mathbf{w} - \mathbf{x}\|$$

Proof. π_U is linear by property (a) of Definition 6.1. The linearity of $\pi_U^\perp = I - \pi_U$ is then immediate.

The values for $\pi_U(\mathbf{x})$ and $\pi_U^\perp(\mathbf{x})$ follow trivially from Definition 6.6 and (*) in Theorem 6.7.

Finally, let $\mathbf{w} \in V$, and let $U = \text{Span}\{\mathbf{w}_1, \dots, \mathbf{w}_n\}$ be spanned by orthonormal vectors: then we have a calculation very like part of the proof of Gram–Schmidt:

$$\begin{aligned} \pi_U^\perp(\mathbf{w}) = \mathbf{w} - \sum_{j=1}^n \langle \mathbf{w}, \mathbf{w}_j \rangle \mathbf{w}_j &\implies \langle \mathbf{w}_i, \pi_U^\perp(\mathbf{w}) \rangle = \langle \mathbf{w}_i, \mathbf{w} \rangle - \sum_{j=1}^n \overline{\langle \mathbf{w}, \mathbf{w}_j \rangle} \langle \mathbf{w}_i, \mathbf{w}_j \rangle \\ &= \langle \mathbf{w}_i, \mathbf{w} \rangle - \overline{\langle \mathbf{w}, \mathbf{w}_i \rangle} \langle \mathbf{w}_i, \mathbf{w}_i \rangle = 0 \end{aligned}$$

Since $\pi_U^\perp(\mathbf{w})$ is orthogonal to a basis of U , it must lie in $U^\perp$.

We can now decompose any vector into U - and $U^\perp$ -parts: $\mathbf{x} = \pi_U(\mathbf{x}) + \pi_U^\perp(\mathbf{x})$. Since $U \cap U^\perp = \{\mathbf{0}\}$, we see that $U \oplus U^\perp = V$.

Finally, use Pythagoras' (Exercise 6.1.3.) to observe that

$$\|\mathbf{w} - \mathbf{x}\|^2 = \|\mathbf{w} - \pi_U(\mathbf{x}) - \pi_U^\perp(\mathbf{x})\|^2 = \|\mathbf{w} - \pi_U(\mathbf{x})\|^2 + \|\pi_U^\perp(\mathbf{x})\|^2$$

whence

$$\|\mathbf{w} - \mathbf{x}\| \geq \|\mathbf{w} - \pi_U(\mathbf{x})\|$$

■

The Theorem also (finally) shows that π_U is independent of the basis of U used in its definition. We already know that $\pi_U(\mathbf{w}) = \mathbf{w}$ whenever $\mathbf{w} \in U$ was basis-independent. Since $V = U \oplus U^\perp$, and $\pi_U(U^\perp) = \mathbf{0}$, the function is now well-defined on all of V independent of choice.

Warning: the Theorem *only* applies when $\dim U < \infty$ since the proof depends crucially on being able to define π_U for which we require a *finite* orthogonal basis. Indeed there are examples where $V \neq U \oplus U^\perp$.

The final property is often known as the *least-squares* property of projections since one is minimizing distance: for the standard inner product on $\mathbb{R}^n$, this amounts to minimizing the distance squared, which is itself a sum of squares. The method of least-squares curve-fitting in statistics can be rephrased as a linear algebra problem involving projections.

Exercises. 6.2.1. Apply the Gram–Schmidt algorithm to each set S and thus obtain an orthogonal basis for $\text{Span } S$, then obtain the co-ordinate representation of the given vector with respect to your basis.

(a) $S = \left\{ \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \right\} \subseteq \mathbb{R}^3$ (standard inner product), $\mathbf{x} = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$

(b) $S = \left\{ \begin{pmatrix} 3 & 5 \\ -1 & 1 \end{pmatrix}, \begin{pmatrix} -1 & 9 \\ 5 & -1 \end{pmatrix}, \begin{pmatrix} 7 & -17 \\ 2 & -6 \end{pmatrix} \right\} \subseteq M_2(\mathbb{R})$ (Frobenius product), $X = \begin{pmatrix} -1 & 27 \\ -4 & 8 \end{pmatrix}$

(c) $S = \{\sin t, \cos t, 1, t\}$ with $\langle f, g \rangle = \int_0^\pi f(t)g(t) dt$ and $h(t) = 2t + 1$

6.2.2. Consider the set $\{\mathbf{w}_1, \mathbf{w}_2\} = \left\{ \begin{pmatrix} 1 \\ 0 \\ 3 \end{pmatrix}, \begin{pmatrix} 2 \\ -1 \\ 0 \end{pmatrix} \right\}$ and let $U = \text{Span}\{\mathbf{w}_1, \mathbf{w}_2\} \leq \mathbb{R}^3$ with the usual inner product. Find the projection of $\mathbf{x} = \begin{pmatrix} 3 \\ -1 \\ -2 \end{pmatrix}$ onto U .

6.2.3. Find the orthogonal complement to $U = \text{Span}\{1 + x, 2 - x^3\} \leq P_3(\mathbb{R})$ with respect to the inner product $\langle f, g \rangle = \int_0^1 f(t)g(t) dt$

6.2.4. Prove Lemma 6.14.

6.2.5. Let V be a finite-dimensional inner product space with orthonormal basis $\beta = \{\mathbf{w}_1, \dots, \mathbf{w}_n\}$ and let $L \in \mathcal{L}(V)$. Prove that the matrix of L with respect to β has jk^{th} entry $\langle L(\mathbf{w}_k), \mathbf{w}_j \rangle$.
(Hint: see Theorem 6.7)

6.2.6. (a) Suppose U is a subspace of an inner product space V . Prove that $U \subseteq (U^\perp)^\perp$.
(Theorem 6.17 says this is equality when U is finite-dimensional)

(b) Let ℓ^2 be the set of square-summable sequences of real numbers (see page 5). Consider the sequences $\mathbf{w}_1, \mathbf{w}_2, \mathbf{w}_3, \dots$, where $\mathbf{w}_j$ is the zero sequence, except for a single 1 in the j th entry. For instance,

$$\mathbf{w}_4 = (0, 0, 0, 1, 0, 0, 0, \dots)$$

i. Let $U = \text{Span}\{\mathbf{w}_j : j \in \mathbb{N}\}$. Prove that $U^\perp$ contains only the zero sequence.

ii. Show that the sequence $(s_n) = (\frac{1}{n})$ lies in ℓ^2 , but does not lie in U ; thus conclude that U is a proper subset of $(U^\perp)^\perp$.

6.2.7. Recall Exercise 6.1.4.: $S = \left\{ \frac{1}{\sqrt{2\pi}} e^{imx} : m \in \mathbb{Z} \right\}$ is orthonormal with respect to $\langle f, g \rangle = \int_{-\pi}^\pi f(t) \overline{g(t)} dt$. The *Fourier series* of a function f is its orthogonal projection onto $\text{Span } S$:

$$\mathcal{F}(f) := \sum_{m=-\infty}^{\infty} \left\langle f, \frac{1}{\sqrt{2\pi}} e^{imx} \right\rangle \frac{1}{\sqrt{2\pi}} e^{imx} = \frac{1}{2\pi} \sum_{m=0}^{\infty} \left(\int_{-\pi}^\pi f(x) e^{-imx} dx \right) e^{imx}$$

Explicitly compute the integrals $\int_{-\pi}^\pi x e^{-imx} dx$ for each $m \in \mathbb{Z}$ and hence show that the Fourier series expression for the function $f(x) = x$ is

$$\mathcal{F}(x) = \sum_{m=1}^{\infty} \frac{2(-1)^{m+1}}{m} \sin mx$$

Sketch a few of the Fourier approximations (sum up to $m = 5$ or $7 \dots$) and observe, when extended to $\mathbb{R}$, how they approximate a discontinuous periodic function.

6.3 The Adjoint of a Linear Operator

For any linear operator on an inner product space, we can describe a related operator.

Definition 6.18. Let V be an inner product space and let $T \in \mathcal{L}(V)$. The *adjoint* of T is a function $T^* : V \rightarrow V$ (L -star) satisfying

$$\forall \mathbf{x}, \mathbf{y} \in V, \quad \langle T^*(\mathbf{x}), \mathbf{y} \rangle = \langle \mathbf{x}, T(\mathbf{y}) \rangle$$

There is much to unpack here, including the usual obvious questions.

Existence Does every linear map have an adjoint?

Uniqueness We wrote ‘The adjoint...’: is this legitimate?

Linearity Is T^* a linear map?

Computation How do we compute T^* given T ?

It turns out that the first question is the trickiest: it will turn out to be true *provided* V is finite-dimensional, though it is possible for a linear transformation of an infinite-dimensional inner product space to have no adjoint. The remaining questions all have nice answers: we’ll prove uniqueness and linearity shortly; for now we assume these and compute an example or two. First we note a simple corollary of Theorem 6.7 as proved in Exercise 6.2.5..

Corollary 6.19. If $T \in \mathcal{L}(V)$ where $\beta = \{\mathbf{w}_1, \dots, \mathbf{w}_n\}$ is an orthonormal basis of V , then the matrix $A = [T]_\beta$ of T with respect to β has jk^{th} entry

$$A_{jk} = \langle T(\mathbf{w}_j), \mathbf{w}_k \rangle$$

Examples 6.20. 1. Let $T = L_A \in \mathcal{L}(\mathbb{R}^2)$ with respect to the standard inner product, where $A = \begin{pmatrix} 3 & 2 \\ -1 & 1 \end{pmatrix}$. Considering the standard orthonormal basis $\{\mathbf{i}, \mathbf{j}\}$, we see that

$$\begin{aligned} \langle T^*(\mathbf{i}), \mathbf{i} \rangle &= \langle \mathbf{i}, T(\mathbf{i}) \rangle = 3 & \langle T^*(\mathbf{j}), \mathbf{i} \rangle &= \langle \mathbf{j}, T(\mathbf{i}) \rangle = -1 \\ \langle T^*(\mathbf{i}), \mathbf{j} \rangle &= \langle \mathbf{i}, T(\mathbf{j}) \rangle = 2 & \langle T^*(\mathbf{j}), \mathbf{j} \rangle &= \langle \mathbf{j}, T(\mathbf{j}) \rangle = 1 \end{aligned}$$

Since T^* is linear, we see that we have computed its matrix with respect to $\{\mathbf{i}, \mathbf{j}\}$: thus

$$T^* = L_B \quad \text{where} \quad B = \begin{pmatrix} 3 & -1 \\ 2 & 1 \end{pmatrix}$$

Note that $B = A^T$ is simply the *transpose* of A .

2. More generally, If $T = L_A \in \mathcal{L}(\mathbb{C}^n)$ with the standard inner product, then, with respect to the standard orthonormal basis $\{\mathbf{e}_1, \dots, \mathbf{e}_n\}$, we see that the matrix B of T^* has jk^{th} entry

$$B_{jk} = \langle T^*(\mathbf{e}_j), \mathbf{e}_k \rangle = \langle \mathbf{e}_j, T(\mathbf{e}_k) \rangle = \overline{\langle T(\mathbf{e}_k), \mathbf{e}_j \rangle} = \overline{A_{kj}}$$

We conclude that the matrix of T^* is the *conjugate transpose* of that of T : $B = A^* = \overline{A}^T$.

Indeed the second example can be generalized very easily with *exactly* the same proof:

Lemma 6.21. If $\beta = \{\mathbf{w}_1, \dots, \mathbf{w}_n\}$ is an orthonormal basis of V and $T \in \mathcal{L}(V)$ has matrix $A = [T]_\beta$, then the matrix of the adjoint T^* with respect to β is the conjugate transpose A^* .

Example 6.22. Let $V = P_2(\mathbb{R})$ be equipped with an inner product defined so that $\epsilon = \{1, x, x^2\}$ is an orthonormal basis. Also let $T = \frac{d}{dx}$. Then the matrix of T is

$$[T]_\epsilon = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{pmatrix}$$

It follows that the matrix of the adjoint is the transpose:

$$[T]_\epsilon^* = \begin{pmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 2 & 0 \end{pmatrix}$$

The adjoint is therefore the linear map defined by

$$T^*(a + bx + cx^2) = ax + 2bx^2$$

To spice things up a bit, consider an alternative inner product, where $\beta = \{1, x - 1, x^2\}$ is orthonormal. With respect to this basis, we have

$$[T(1)]_\beta = [0]_\beta = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}, \quad [T(x - 1)]_\beta = [1]_\beta = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad [T(x^2)]_\beta = [2x]_\beta = \begin{pmatrix} 2 \\ 2 \\ 0 \end{pmatrix}$$

It follows that the adjoint has matrix

$$[T]_\beta^* = \begin{pmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 2 & 2 & 0 \end{pmatrix}$$

In particular,

$$T^*(1) = (x - 1) + 2(x^2) = 2x^2 + x - 1,$$

$$T^*(x) = T^*(x - 1 + 1) = 2x^2 + (2x^2 + x - 1) = 4x^2 + x - 1$$

$$T^*(x^2) = 0$$

Notice that this is a very different linear map from above, which is to be expected: the inner product spaces are different!

Tidying up the Definition

Lemma 6.23. *If a linear map has an adjoint, then it is linear and unique.*

Proof. Suppose T^* and S^* are adjoints of T . Then

$$\langle T^*(\mathbf{x}), \mathbf{y} \rangle = \langle \mathbf{x}, T(\mathbf{y}) \rangle = \langle S^*(\mathbf{y}), \mathbf{z} \rangle$$

Since this holds for all $\mathbf{y}$, Lemma 6.3 part 5 says that $\forall \mathbf{x}, T^*(\mathbf{x}) = S^*(\mathbf{x})$, whence $T^* = S^*$.

It is similarly easy to check that

$$\langle T^*(\lambda \mathbf{x} + \mathbf{y}), \mathbf{z} \rangle = \langle \lambda T^*(\mathbf{x}) + T^*(\mathbf{y}), \mathbf{z} \rangle$$

Another appeal to the lemma establishes the linearity of T^* . ■

Before addressing existence, we consider a natural isomorphism between a finite-dimensional inner product space and its *dual-space*. By Corollary 6.12, such a V has an orthonormal basis $\beta = \{\mathbf{w}_1, \dots, \mathbf{w}_n\}$. Let $g : V \rightarrow \mathbb{F}$ be linear ($g \in V^*$ is a member of the *dual space*) and define

$$\mathbf{y} := \sum_{j=1}^n \overline{g(\mathbf{w}_j)} \mathbf{w}_j$$

Certainly $\mathbf{x} \mapsto \langle \mathbf{x}, \mathbf{y} \rangle$ is a linear map and thus an element of V^* . Moreover,

$$\langle \mathbf{w}_k, \mathbf{y} \rangle = \left\langle \mathbf{w}_k, \sum_{j=1}^n \overline{g(\mathbf{w}_j)} \mathbf{w}_j \right\rangle = \sum_{j=1}^n g(\mathbf{w}_j) \langle \mathbf{w}_k, \mathbf{w}_j \rangle = g(\mathbf{w}_k)$$

Since these linear maps are equal on a basis, they are identical. We conclude:

Theorem 6.24. *If $g : V \rightarrow \mathbb{F}$ is a linear map on a finite-dimensional inner product space, then there exists a unique $\mathbf{y} \in V$ such that $g(\mathbf{x}) = \langle \mathbf{x}, \mathbf{y} \rangle$. Moreover, $\phi : V \rightarrow V^* : \mathbf{y} \mapsto g$ is an isomorphism.*

Theorem 6.25 (Existence of the adjoint). *Every linear operator on a finite-dimensional vector space has an adjoint.*

Proof. Let $T \in \mathcal{L}(V)$ and suppose $\mathbf{z} \in V$ is given: our goal is to define $T^*(\mathbf{z})$ and to check that it satisfies the correct properties.

Define $g(\mathbf{x}) = \langle T(\mathbf{x}), \mathbf{z} \rangle$. It is easy to check that this is linear, whence $g \in V^*$. By Theorem 6.24, $\exists$ a unique $\mathbf{y} \in V$ such that

$$g(\mathbf{x}) = \langle \mathbf{x}, \mathbf{y} \rangle$$

Define $T^* : V \rightarrow V$ by $T^*(\mathbf{z}) = \mathbf{y}$. It is immediate that

$$\langle T^*(\mathbf{z}), \mathbf{x} \rangle = \langle \mathbf{y}, \mathbf{x} \rangle = \overline{g(\mathbf{x})} = \langle \mathbf{z}, T(\mathbf{x}) \rangle$$

whence T^* is the adjoint of T . ■

It is important to observe the restriction to a finite-dimensional inner product space in these theorems. The infinite-dimensional case is much more complicated; in such a situation there might exist linear operators without an adjoint. However, there is a class of inner product spaces known as *Hilbert spaces* which satisfy an analysis-style completeness property; the common examples are infinite-dimensional and are built on the ℓ^2 and L^2 inner products mentioned previously. A more general version of Theorem 6.24 applies in these cases and is known as the *Riesz representation theorem*: it guarantees the existence of the adjoint of every *bounded* linear operator on a Hilbert space.

Projections and Least Squares Problems

The conjugate transpose of a matrix provides a rapid technique for computing projections in the inner product space $\mathbb{F}^m$ equipped with the standard inner product. To see how this works, we first need an easy result:

Theorem 6.26. Let $A \in M_{m \times n}(\mathbb{F})$.

1. $\forall \mathbf{x} \in \mathbb{F}^n, \mathbf{y} \in \mathbb{F}^m$, we have $\langle A\mathbf{x}, \mathbf{y} \rangle = \langle \mathbf{x}, A^*\mathbf{y} \rangle$, where the first inner product lies in $\mathbb{F}^m$ and the second in $\mathbb{F}^n$.
2. $\text{rank}(A^*A) = \text{rank } A$, so that $A^*A \in M_n(\mathbb{F})$ is invertible if and only if $\text{rank } A = n$.

Proof. 1. This follows trivially from the fact that the standard inner product is $\langle \mathbf{w}, \mathbf{w} \rangle = \mathbf{w}^* \mathbf{w}$.

2. $A^*A\mathbf{x} = \mathbf{0} \implies \langle A^*A\mathbf{x}, \mathbf{x} \rangle = \langle A\mathbf{x}, A\mathbf{x} \rangle = 0 \implies A\mathbf{x} = \mathbf{0}$. Conversely $A\mathbf{x} = \mathbf{0} \implies A^*A\mathbf{x} = \mathbf{0}$. Thus A^*A and A have the same nullspace. By the rank-nullity theorem, n is the sum of the rank and nullity of both, whence both have the same rank. ■

Consider the following problem. Let $A \in M_{m \times n}(\mathbb{F})$ and $\mathbf{y} \in \mathbb{F}^m$ be given and define $U = C(A)$ to be the column space of A . Our goal is to find a vector $\mathbf{x}_0 \in \mathbb{F}^n$ such that

$$\forall \mathbf{x} \in \mathbb{F}^n, \quad \|\mathbf{y} - A\mathbf{x}_0\| \leq \|\mathbf{y} - A\mathbf{x}\| \quad (*)$$

We know from Theorem 6.17 that $\mathbf{x}_0$ minimizes the function $\|\mathbf{y} - A\mathbf{x}\|$ precisely when $A\mathbf{x}_0 = \pi_U(\mathbf{y})$. But this is precisely when

$$\mathbf{y} - A\mathbf{x}_0 \in U^\perp \iff \forall \mathbf{w} \in U, \langle \mathbf{y} - A\mathbf{x}_0, \mathbf{w} \rangle = 0 \iff \forall \mathbf{z} \in \mathbb{F}^n, \langle \mathbf{y} - A\mathbf{x}_0, A\mathbf{z} \rangle = 0$$

since U is the column space of A . However, this is if and only if

$$\forall \mathbf{z} \in \mathbb{F}^n, \langle A^*A\mathbf{x}_0 - A^*\mathbf{y}, \mathbf{z} \rangle = 0 \iff A^*A\mathbf{x}_0 = A^*\mathbf{y}$$

Since $\text{rank}(A^*A) = \text{rank } A = \text{rank } A^*$, we see that A^*A and A^* have the same column space, whence this equation always has a solution. To summarize:

Theorem 6.27. Suppose $A \in M_{m \times n}(\mathbb{F})$ and $\mathbf{y} \in \mathbb{F}^m$. Then $\exists \mathbf{x}_0 \in \mathbb{F}^n$ such that $A^*A\mathbf{x}_0 = A^*\mathbf{y}$ which moreover satisfies (*).

If $\text{rank } A = n$, then A^*A is invertible, and $\mathbf{x}_0 = (A^*A)^{-1}A^*\mathbf{y}$. Moreover, with respect to the standard basis, $[\pi_U]_e = A(A^*A)^{-1}A^*$.

Example 6.28. Compute the orthogonal projection onto $U = \text{Span} \left\{ \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}, \begin{pmatrix} 3 \\ 1 \\ -2 \end{pmatrix} \right\}$. Taking the columns of A to be the basis of U , we compute

$$\begin{aligned} [\pi_U]_e &= A(A^*A)^{-1}A^* = \begin{pmatrix} 1 & 3 \\ 0 & 1 \\ -1 & -2 \end{pmatrix} \left[\begin{pmatrix} 1 & 0 & -1 \\ 3 & 1 & -2 \end{pmatrix} \begin{pmatrix} 1 & 3 \\ 0 & 1 \\ -1 & -2 \end{pmatrix} \right]^{-1} \begin{pmatrix} 1 & 0 & -1 \\ 3 & 1 & -2 \end{pmatrix} \\ &= \begin{pmatrix} 1 & 3 \\ 0 & 1 \\ -1 & -2 \end{pmatrix} \begin{pmatrix} 2 & 5 \\ 5 & 14 \end{pmatrix}^{-1} \begin{pmatrix} 1 & 0 & -1 \\ 3 & 1 & -2 \end{pmatrix} = \frac{1}{3} \begin{pmatrix} 1 & 3 \\ 0 & 1 \\ -1 & -2 \end{pmatrix} \begin{pmatrix} 14 & -5 \\ -5 & 2 \end{pmatrix} \begin{pmatrix} 1 & 0 & -1 \\ 3 & 1 & -2 \end{pmatrix} \\ &= \frac{1}{3} \begin{pmatrix} 2 & 1 & -1 \\ 1 & 2 & 1 \\ -1 & 1 & 2 \end{pmatrix} \end{aligned}$$

You should check that $\pi_U(\mathbf{w}) = \mathbf{w}$ for any $\mathbf{w} \in U$ and that $\pi_U \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix} = \mathbf{0}$. Moreover, you could easily choose an alternative basis of U , for instance $\left\{ \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} \right\}$; the projection map will be the same!

This process will allow you to compute projections more generally; the problem is that you have to be able to compute adjoints, which is only easy when everything is written with respect to an orthonormal basis.

We now come to a major application. Given a data set of points in the plane $\{(t_j, y_j) : 1 \leq j \leq m\}$ (where $m \geq 2$), we want to find the straight line² (equation $y = c_0 + c_1 t$) which minimizes the sum of the squares of the vertical deviations

$$\sum_{j=1}^m |y_j - c_1 t_j - c_0|^2$$

As such, we call this a *least-squares problem*. By the above discussion, we know how to solve it: define

$$A = \begin{pmatrix} t_1 & 1 \\ \vdots & \vdots \\ t_m & 1 \end{pmatrix} \quad \mathbf{x} = \begin{pmatrix} c_1 \\ c_0 \end{pmatrix} \quad \mathbf{y} = \begin{pmatrix} y_1 \\ \vdots \\ y_m \end{pmatrix} \implies \sum_{j=1}^m |y_j - c_1 t_j - c_0|^2 = \|\mathbf{y} - A\mathbf{x}\|^2$$

The solution is to choose $\mathbf{x}_0 = \begin{pmatrix} c_1 \\ c_0 \end{pmatrix} \in \mathbb{F}^2$ to be any vector satisfying $A^*A\mathbf{x}_0 = A^*\mathbf{y}$: indeed, unless $t_1 = t_2 = \dots = t_m$, we see that $\text{rank } A = 2$, whence there is a unique solution

$$\mathbf{x}_0 = \begin{pmatrix} c_1 \\ c_0 \end{pmatrix} = (A^*A)^{-1}A^*\mathbf{y}$$

We call the straight line $y = c_1 t + c_0$ the *best-fitting least-squares straight line* for the data.

²This is the *linear regression* line from statistics, used to predict y given a value of t .

Example 6.29. Given the data set $\{(0, 1), (1, 1), (2, 0), (3, 2), (4, 2)\}$, we compute

$$A = \begin{pmatrix} 0 & 1 \\ 1 & 1 \\ 2 & 1 \\ 3 & 1 \\ 4 & 1 \end{pmatrix}, \quad \mathbf{y} = \begin{pmatrix} 1 \\ 1 \\ 0 \\ 2 \\ 2 \end{pmatrix} \Rightarrow \mathbf{x}_0 = \begin{pmatrix} c_1 \\ c_0 \end{pmatrix} = (A^* A)^{-1} A^* \mathbf{y} = \begin{pmatrix} 30 & 10 \\ 10 & 5 \end{pmatrix}^{-1} \begin{pmatrix} 15 \\ 6 \end{pmatrix} = \begin{pmatrix} 0.3 \\ 0.6 \end{pmatrix}$$

The regression line therefore has equation $y = 0.3t + 0.6$.

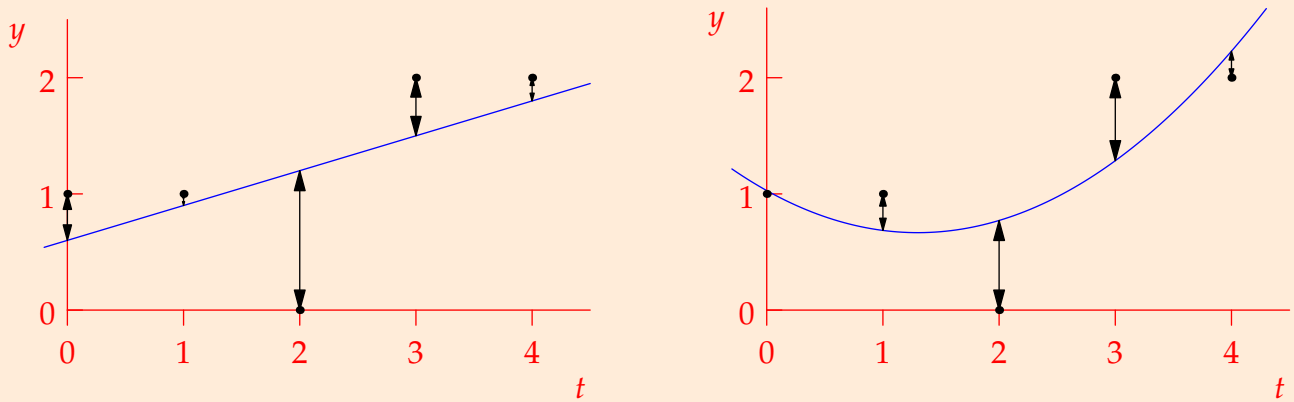
The process can be applied more generally to approximate a data set using many other families of functions. For instance, to find the best-fitting quadratic polynomial $y = c_2 t^2 + c_1 t + c_0$, we'd work with

$$A = \begin{pmatrix} t_1^2 & t_1 & 1 \\ \vdots & \vdots & \vdots \\ t_m^2 & t_m & 1 \end{pmatrix} \quad \mathbf{x} = \begin{pmatrix} c_2 \\ c_1 \\ c_0 \end{pmatrix} \quad \mathbf{y} = \begin{pmatrix} y_1 \\ \vdots \\ y_m \end{pmatrix} \Rightarrow \sum_{j=1}^m |y_j - c_2 t_j^2 - c_1 t_j - c_0|^2 = \|\mathbf{y} - A\mathbf{x}\|^2$$

Provided we have at least three distinct values t_1, t_2, t_3 , the matrix A is guaranteed to have rank 3, and there will be a best-fitting least-squares quadratic: in this case

$$y = \frac{1}{70}(15t^2 - 39t + 72)$$

This curve, and the best-fitting straight line are shown below.



The above process can easily be extended to a degree n polynomial or to other families of functions. For instance, to find the best-fitting least-squares curve of the form $y = ae^t + b$ to the data, we would apply the method to

$$A = \begin{pmatrix} e^{t_1} & 1 \\ \vdots & \vdots \\ e^{t_m} & 1 \end{pmatrix} \quad \mathbf{x} = \begin{pmatrix} a \\ b \end{pmatrix} \quad \mathbf{y} = \begin{pmatrix} y_1 \\ \vdots \\ y_m \end{pmatrix} \Rightarrow \sum_{j=1}^m |y_j - ae^{t_j} - b|^2 = \|\mathbf{y} - A\mathbf{x}\|^2$$

Exercises. 6.3.1. For each of the inner product spaces V and linear transformations $g : V \rightarrow \mathbb{F}$, find a vector $\mathbf{y} \in V$ such that $g(\mathbf{x}) = \langle \mathbf{x}, \mathbf{y} \rangle$ for all $\mathbf{x} \in V$.

- (a) $V = \mathbb{R}^3$ with the standard inner product, and $g \begin{pmatrix} x \\ y \\ z \end{pmatrix} = x - 2y + 4z$
- (b) $V = \mathbb{C}^2$ with the standard inner product, and $g \begin{pmatrix} z \\ w \end{pmatrix} = iz - 2w$
- (c) $V = P_2(\mathbb{R})$ with the L^2 inner product $\langle f, h \rangle = \int_0^1 f(t)h(t) dt$, and $g(f) = f(0) + f'(1)$

6.3.2. For each inner product space V and linear operator $T \in \mathcal{L}(V)$, evaluate T^* on the given vector.

- (a) $V = \mathbb{R}^2$ with the standard inner product, $T \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 2x+y \\ x-3y \end{pmatrix}$ and $\mathbf{x} = \begin{pmatrix} 3 \\ 5 \end{pmatrix}$
- (b) $V = \mathbb{C}^2$ with the standard inner product, $T \begin{pmatrix} z \\ w \end{pmatrix} = \begin{pmatrix} 2z+iw \\ (1-i)z \end{pmatrix}$ and $\mathbf{x} = \begin{pmatrix} 3-i \\ 1+2i \end{pmatrix}$
- (c) $V = P_1(\mathbb{R})$ with $\langle f, g \rangle = \int_0^1 f(t)g(t) dt$, $T(f) = f' + 3f$ and $f(t) = 4 - 2t$

6.3.3. Let $T(f) = f''$ be a linear transformation of $P_2(\mathbb{R})$ and let $\epsilon = \{1, x, x^2\}$ be the standard basis. Find $T^*(a + bx + cx^2)$:

- (a) With respect to the inner product where ϵ is orthonormal;
- (b) With respect to the inner product $\langle f, g \rangle = \int_{-1}^1 f(t)g(t) dt$.
(Hint: $\{1, x, x^2 - \frac{1}{3}\}$ is orthogonal, so project $T^*(1)$, etc., onto its span and use Theorem 6.7...)

6.3.4. Prove the Fundamental Subspaces Theorem for an inner product space V : let $T \in \mathcal{L}(V)$ have adjoint T^* and let $\mathcal{R}$ and $\mathcal{N}$ denote the range and null space of a linear operator.

- (a) $\mathcal{R}(T^*)^\perp = \mathcal{N}(T)$
- (b) If V is finite dimensional, $\mathcal{R}(T^*) = \mathcal{N}(T)^\perp$

6.3.5. Let V be an inner product space and let $\mathbf{y}, \mathbf{z} \in V$ be fixed. Define $T \in \mathcal{L}(V)$ by

$$T(\mathbf{x}) = \langle \mathbf{x}, \mathbf{y} \rangle \mathbf{z}$$

Show that T^* exists and find an explicit expression for it.

6.3.6. Suppose $A \in M_{m \times n}(\mathbb{F})$ is such that no two columns are identical. Prove that A^*A is diagonal if and only if the columns of A are orthogonal. What additionally would it mean if $A^*A = I$?

6.3.7. Check the calculation for the best-fitting least-squares quadratic in Example 6.29.

6.3.8. Find the best-fitting least-squares linear and quadratic approximations to the data set

$$\{(1, 2), (3, 4), (5, 7), (7, 9), (9, 12)\}$$

6.3.9. (Particularly if you've done statistics/probability) Consider the best-fitting least-squares line with equation $y = ct + d$ through a data set $\{(t_j, y_j) : 1 \leq j \leq m\}$.

- (a) Show that the equations $A^*A\mathbf{x}_0 = A^*\mathbf{y}$ can be written in matrix form

$$\begin{pmatrix} \sum t_j^2 & \sum t_j \\ \sum t_j & m \end{pmatrix} \begin{pmatrix} c \\ d \end{pmatrix} = \begin{pmatrix} \sum t_j y_j \\ \sum y_j \end{pmatrix}$$

- (b) Show that $c = \frac{\text{Cov}(t, y)}{\sigma_t^2}$ and $d = \bar{y} - c\bar{t}$ where $\bar{t} = \frac{1}{m} \sum t_j$, $\bar{y} = \frac{1}{m} \sum y_j$, $\sigma_t^2 = \frac{1}{m} \sum (t_j - \bar{t})^2$ and $\text{Cov}(t, y) = \frac{1}{m} \sum (t_j - \bar{t})(y_j - \bar{y})$ are the means, variance and covariance respectively.

6.4 Normal and Self-Adjoint Operators

We now consider a fundamental question: given a linear operator $T \in \mathcal{L}(V)$, can we find an orthonormal basis of V consisting of eigenvectors of T ? For a general map this is far too much to hope for: indeed the linear maps for which this is possible are very, very special. We start with the definition and the major result.

Definition 6.30. Suppose T is a linear operator on an inner product space V which has an adjoint. We say that T is:

Normal if $TT^* = T^*T$, where juxtaposition is composition of functions;

Self-adjoint if $T^* = T$.

The corresponding definitions for real or complex square matrices are identical.

A few notes:

- A real matrix $A \in M_n(\mathbb{R})$ is self-adjoint if and only if it is *symmetric*: $A^T = A$.
- If T is self-adjoint then it is normal, but the converse is false. For instance, every non-zero real *skew-symmetric* matrix is normal but *not* self-adjoint:

$$A^T = -A \implies AA^T = -A^2 = A^T A$$

More generally, the same applies to *skew-hermitian* matrices $A^* = -A$.

Example 6.31. The matrix $A = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ is skew-symmetric. Observe that $A^T = -A$ so that A is not self-adjoint. However

$$AA^T = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = A^T A$$

so that A is normal.

The main purpose of this section is to prove the following important result. It comes in two flavors, depending on whether an inner product space is real or complex.

Theorem 6.32 (Spectral Theorem). Let T be a linear operator on a finite-dimensional inner product space V over $\mathbb{F}$.

Complex case $\mathbb{F} = \mathbb{C}$: V has an orthonormal basis of eigenvectors of T if and only if T is normal.

Real case $\mathbb{F} = \mathbb{R}$: V has an orthonormal basis of eigenvectors of T if and only if T is self-adjoint.

In particular, every such linear operator is diagonalizable.

Before starting this work, we consider several easy examples.

Examples 6.33. 1. The linear map $T = L_A$ where $A = \begin{pmatrix} 6 & 3 \\ 3 & -2 \end{pmatrix}$ is self-adjoint (and thus normal) in the standard inner product space $\mathbb{R}^2$. To diagonalize, we find the eigenvalues and eigenvectors:

Characteristic polynomial: $p(t) = (6 - t)(-2 - t) - 9 = t^2 - 4t - 21 = (t - 7)(t + 3)$

Eigenvalues: $t_1 = 7, t_2 = -3$

Eigenvectors (normalized): $\mathbf{w}_1 = \frac{1}{\sqrt{10}} \begin{pmatrix} 3 \\ 1 \end{pmatrix}, \mathbf{w}_2 = \frac{1}{\sqrt{10}} \begin{pmatrix} -1 \\ 3 \end{pmatrix}$

The basis $\beta = \{\mathbf{w}_1, \mathbf{w}_2\}$ is orthonormal, with respect to which $[T]_\beta = \begin{pmatrix} 7 & 0 \\ 0 & -3 \end{pmatrix}$ is diagonal.

2. The linear map $T = L_A$ where $A = \begin{pmatrix} 1 & 3 \\ 0 & -2 \end{pmatrix}$ is not self-adjoint, and not normal either.^a It is diagonalizable, but only with respect to a basis of eigenvectors, which a quick calculation will confirm to be, up to order and scaling,

$$\beta = \left\{ \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 3 \\ -1 \end{pmatrix} \right\} \rightsquigarrow [T]_\beta = \begin{pmatrix} 1 & 0 \\ 0 & -2 \end{pmatrix}$$

Of course this basis is not orthogonal, in accordance with Theorem 6.32.

3. We revisit Example 6.31 and consider the linear maps defined by multiplication by $A = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ on both $\mathbb{C}^2$ and $\mathbb{R}^2$: since A is normal but not self-adjoint, we'll see how the field really matters when it comes to the spectral theorem.

First the complex case: according to the spectral theorem, $T = L_A \in \mathcal{L}(\mathbb{C}^2)$ is normal and thus should be diagonalizable with respect to an orthonormal basis of eigenvectors. Here are the details.

Characteristic polynomial: $p(t) = t^2 + 1 = (t - i)(t + i)$

Eigenvalues: $t_1 = i, t_2 = -i$

Eigenvectors (normalized): $\mathbf{w}_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} i \\ 1 \end{pmatrix}, \mathbf{w}_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} -i \\ 1 \end{pmatrix}$

Certainly $\langle \mathbf{w}_1, \mathbf{w}_2 \rangle = 0$ so that $\beta = \{\mathbf{w}_1, \mathbf{w}_2\}$ is orthonormal; moreover $[T]_\beta = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}$ is diagonal.

Now for the real case: the linear map $S = L_A \in \mathcal{L}(\mathbb{R}^2)$ is not self-adjoint and thus should not be diagonalizable with respect to an orthonormal basis of eigenvectors. In one sense this is trivial: S is rotation by 90° counter-clockwise around the origin, so it has no eigenvectors! Indeed the characteristic polynomial has no roots in $\mathbb{R}$: there are no eigenvalues. For completeness, it is worth considering S with respect to any orthonormal basis of $\mathbb{R}^2$: with a little thought, you should believe that all such bases have the form

$$\beta = \{\mathbf{w}, \mathbf{u}\} = \left\{ \begin{pmatrix} \cos \theta \\ \sin \theta \end{pmatrix}, \begin{pmatrix} -\sin \theta \\ \cos \theta \end{pmatrix} \right\} \quad \text{or} \quad \gamma = \{\mathbf{w}, -\mathbf{u}\} \quad \text{for some } \theta.$$

Observe that $S(\mathbf{w}) = \mathbf{u}$ and $S(\mathbf{u}) = -\mathbf{w}$. It follows that

$$[S]_\beta = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = A, \quad [S]_\gamma = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} = -A$$

The matrix of S is essentially the same with respect to *any* orthonormal basis! In particular, there is no orthonormal basis with respect to which the matrix of S is diagonal.

^a $AA^* = \begin{pmatrix} 1 & 3 \\ 0 & -2 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 3 & -2 \end{pmatrix} = \begin{pmatrix} 10 & -6 \\ -6 & 4 \end{pmatrix} \neq \begin{pmatrix} 1 & 3 \\ 3 & 13 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 3 & -2 \end{pmatrix} \begin{pmatrix} 1 & 3 \\ 0 & -2 \end{pmatrix} = A^*A$

We begin the proof of the spectral theorem by stating some basic properties of normal operators.

Lemma 6.34. *Let T be normal on V . Then:*

1. $\forall \mathbf{x} \in V, \|T(\mathbf{x})\| = \|T^*(\mathbf{x})\|$;
2. $T - kI$ is normal for any $k \in \mathbb{F}$;
3. $T(\mathbf{x}) = \lambda \mathbf{x} \iff T^*(\mathbf{x}) = \bar{\lambda} \mathbf{x}$ so that T and T^* have the same eigenvectors and conjugate eigenvalues. If T is additionally self-adjoint, then every eigenvalue is real.
4. Distinct eigenvalues of T have orthogonal eigenvectors.

Note that the theorem applies to self-adjoint operators as a special case.

Proof. 1. $\|T(\mathbf{x})\|^2 = \langle T(\mathbf{x}), T(\mathbf{x}) \rangle = \langle T^*T(\mathbf{x}), \mathbf{x} \rangle = \langle TT^*(\mathbf{x}), \mathbf{x} \rangle = \langle T^*(\mathbf{x}), T^*(\mathbf{x}) \rangle = \|T^*(\mathbf{x})\|^2$.

2. $\langle \mathbf{x}, (T - kI)(\mathbf{y}) \rangle = \langle \mathbf{x}, T(\mathbf{y}) \rangle - k \langle \mathbf{x}, \mathbf{y} \rangle = \langle T^*(\mathbf{x}), \mathbf{y} \rangle - \langle \bar{k} \mathbf{x}, \mathbf{y} \rangle = \langle (T^* - \bar{k}I)(\mathbf{x}), \mathbf{y} \rangle$ shows that $T - kI$ has adjoint $T^* - \bar{k}I$. It is trivial to check that these commute.

3. $T(\mathbf{x}) = \lambda \mathbf{x} \iff \|(T - \lambda I)(\mathbf{x})\| = 0 \xrightarrow[1 \& 2]{\text{parts}} \|(T^* - \bar{\lambda}I)(\mathbf{x})\| = 0 \iff T^*(\mathbf{x}) = \bar{\lambda} \mathbf{x}$.
Moreover, if $T^* = T$, then $\bar{\lambda} = \lambda$.

4. Suppose $T(\mathbf{x}) = \lambda \mathbf{x}$ and $T(\mathbf{y}) = \mu \mathbf{y}$ where $\lambda \neq \mu$. By part 3,

$$\begin{aligned} \langle T(\mathbf{x}), \mathbf{y} \rangle &= \langle \lambda \mathbf{x}, \mathbf{y} \rangle = \lambda \langle \mathbf{x}, \mathbf{y} \rangle \\ &\parallel \\ \langle \mathbf{x}, T^*(\mathbf{y}) \rangle &= \langle \mathbf{x}, \bar{\mu} \mathbf{y} \rangle = \bar{\mu} \langle \mathbf{x}, \mathbf{y} \rangle \end{aligned}$$

This is a contradiction unless $\langle \mathbf{x}, \mathbf{y} \rangle = 0$. ■

Example 6.35. Consider $T = L_A$ where $A = \begin{pmatrix} 0 & 3+4i \\ -3+4i & 0 \end{pmatrix}$. This is skew-hermitian ($A^* = -A$) and thus normal but not self-adjoint. To find its eigenpairs, we solve

$$p(t) = t^2 - (3+4i)(-3+4i) = t^2 + 25 = 0 \implies t = \pm 5i$$

The eigenvectors can then be found:

$$\lambda = 5i \implies \begin{pmatrix} -5i & 3+4i \\ -3+4i & -5i \end{pmatrix} \mathbf{w} = \mathbf{0} \implies \text{choose } \mathbf{w}_1 = \begin{pmatrix} 3+4i \\ 5i \end{pmatrix}$$

while the other eigenpair is seen to be $\bar{\lambda}, \mathbf{w}_2 = \begin{pmatrix} 3+4i \\ -5i \end{pmatrix}$. Observe that these eigenvectors are orthogonal:

$$\langle \mathbf{w}_1, \mathbf{w}_2 \rangle = \mathbf{w}_2^* \mathbf{w}_1 = (\overline{3+4i})(3+4i) + (\overline{-5i})(5i) = 25 - 25 = 0$$

Note also that the adjoint $A^* = -A$ has the same eigenvectors but with conjugate eigenvalues:

$$A^* \mathbf{w} = -A \mathbf{w} = -\lambda \mathbf{w} = \bar{\lambda} \mathbf{w}$$

As a consequence of Lemma 6.34, any *diagonalizable* normal or self-adjoint operators necessarily have an orthogonal eigenbasis. The hard part of the spectral theorem is proving that such operators are diagonalizable in the first place! This can be done in one shot, but first we develop a useful and interesting result.

Lemma 6.36 (Schur). Suppose $T \in \mathcal{L}(V)$ where $\dim V < \infty$. If the characteristic polynomial of T splits,^a then there exists an orthonormal basis β of V such that $[T]_\beta$ is upper-triangular.

^aCompletely factorizes into linear factors over $\mathbb{F}$: this is guaranteed over $\mathbb{C}$ by the Fundamental Theorem of Algebra.

Schur's Lemma is stronger than the spectral theorem since it applies to more (non-normal) linear operators, however its conclusion is weaker ($[T]_\beta$ upper-triangular rather than diagonal). Every example of the spectral theorem we've seen is therefore an example of Schur's Lemma. Example 6.33.2 provides another trivial example, since the matrix A is already upper triangular. Here is another:

Example 6.37. Consider $T(f) = 2f'(x) + xf(1)$ as a linear map $T \in \mathcal{L}(P_1(\mathbb{R}))$ with respect to the inner product $\langle f, g \rangle = \int_0^1 f(t)g(t) dt$. We have

$$T(a + bx) = 2b + (a + b)x$$

If $[T]_\beta$ is to be upper-triangular, then the first vector in β must be an eigenvector of T . It is easily checked that $f_1 = 1 + x$ is such with eigenvalue 2. To find a basis satisfying Schur's Lemma, we need only find f_2 orthogonal to this and then normalize. This can be done by brute force since the problem is small, but for the sake of practice we apply Gram-Schmidt to the polynomial 1:

$$1 - \frac{\langle 1, 1+x \rangle}{\|1+x\|^2} (1+x) = 1 - \frac{1 + \frac{1}{2}}{1 + 1 + \frac{1}{3}} (1+x) = \frac{1}{14} (5 - 9x) \implies f_2 = 5 - 9x$$

Indeed we obtain an upper-triangular matrix for T :

$$T(f_2) = -18 - 4x = -13(1+x) - (5-9x) = -13f_1 - f_2 \implies [T]_{\{f_1, f_2\}} = \begin{pmatrix} 2 & -13 \\ 0 & -1 \end{pmatrix}$$

We can also work with the corresponding orthonormal basis as posited in the theorem, though the matrix is messier:

$$\beta = \{g_1, g_2\} = \left\{ \sqrt{\frac{3}{7}}(1+x), \frac{1}{\sqrt{7}}(5-9x) \right\} \implies [T]_\beta = \begin{pmatrix} 2 & -\frac{13}{\sqrt{3}} \\ 0 & -1 \end{pmatrix}$$

Alternatively, we could have started with the other eigenvector $h_1 = 2 - x$: an orthogonal vector to this is $h_2 = 4 - 9x$, with respect to which

$$[T]_{\{h_1, h_2\}} = \begin{pmatrix} -1 & -13 \\ 0 & 2 \end{pmatrix}$$

In both cases the eigenvalues are down the diagonal, as must be for an upper-triangular matrix!

Unfortunately the tactic of extending a single eigenvector to a basis doesn't work so well when $\dim V \geq 3$. Instead we appeal to the more general idea of an invariant subspace. Recall that $U \leq V$ being T -invariant means that $T(U) \leq U$: moreover, if γ is a basis of U extended to a basis $\beta = \gamma \cup \eta$ of V , then the matrix of T with respect to β has block form

$$[T]_{\beta} = \left(\begin{array}{c|c} A & B \\ \hline 0 & C \end{array} \right)$$

The core of the proof is an induction argument: if $\dim U = \dim V - 1$ and A is upper-triangular, then $[T]_{\beta}$ is upper-triangular.³

Proof of Schur's Lemma. If $\dim V = 1$, then the result is trivial: all linear maps have the form $T(\mathbf{x}) = \lambda \mathbf{x}$ for some λ .

Now suppose the result is true for all inner product spaces with dimension n . Suppose also that $\dim V = n + 1$, and that the characteristic polynomial of $T \in \mathcal{L}(V)$ splits. It follows that T has at least one eigenpair $(\mu, \mathbf{w})$. Moreover, since V is finite-dimensional, T has an adjoint. Indeed,

$$\begin{aligned} \forall \mathbf{x} \in V, 0 = \langle T(\mathbf{w}) - \mu \mathbf{w}, \mathbf{x} \rangle &= \langle \mathbf{w}, T^*(\mathbf{x}) - \bar{\mu} \mathbf{x} \rangle \implies \mathcal{R}(T^* - \bar{\mu}I) \leq \text{Span}\{\mathbf{w}\}^{\perp} \\ &\implies \text{rank}(T^* - \bar{\mu}I) \leq n < \dim V \end{aligned}$$

By the Rank–Nullity Theorem, $T^* - \bar{\mu}I$ has a non-trivial null space: otherwise said, T^* has an eigenvector $\mathbf{u}$ (assume unit length) with eigenvalue $\bar{\mu}$. Let $W = \text{Span}\{\mathbf{u}\}$ and let $\mathbf{y} \in W^{\perp}$. Then

$$\langle T(\mathbf{y}), \mathbf{u} \rangle = \langle \mathbf{y}, T^*(\mathbf{u}) \rangle = \bar{\mu} \langle \mathbf{y}, \mathbf{u} \rangle = 0 \implies T(W^{\perp}) \leq W^{\perp}$$

whence $W^{\perp}$ is T -invariant. Recall chapter 5: the characteristic polynomial of the restriction $p_{W^{\perp}}(t)$ divides that of T on V : thus $p_{W^{\perp}}(t)$ splits. By the induction hypothesis, $W^{\perp}$ has a basis γ with respect to which $[T|_{W^{\perp}}]_{\gamma}$ is upper-triangular. Let $\beta = \gamma \cup \{\mathbf{u}\}$ to finish the induction and thus the proof. ■

We are now (finally!) able to prove the spectral theorem. First we tackle the real case. Since the proof that follows is a little sneaky, we first give a synopsis. To invoke Schur's Lemma we need to see that the characteristic polynomial of T (an operator on a *real* inner product space) splits over $\mathbb{R}$. The trick is to first transfer to a complex inner product space so we can appeal to the Fundamental Theorem of Algebra. Here are the main steps:

1. Create an associated self-adjoint linear operator S on $\mathbb{C}^n$ with the *same* characteristic polynomial as T .
2. The Fundamental Theorem of Algebra says this splits over $\mathbb{C}$.
3. Self-adjointness says the eigenvalues are *real*.
4. Apply Schur's Lemma for an orthonormal basis.
5. Show that the resulting matrix is diagonal.

Proof of the Spectral Theorem for real inner product spaces. Let $\dim V = n$, T be self-adjoint, choose any orthonormal basis γ of V , let $A = [T]_\gamma$ and define $S = L_A \in \mathcal{L}(\mathbb{C}^n)$ to be multiplication by A . With respect to the standard basis ϵ of $\mathbb{C}^n$, the matrix of S is $[S]_\epsilon = A$. Two facts are immediate:

- The characteristic polynomials of S and T are identical (to that of A).
- S is self-adjoint and thus (Lemma 6.34) has real eigenvalues.

By the Fundamental Theorem of Algebra, the characteristic polynomial splits *over* $\mathbb{C}$

$$p(t) = (\lambda_1 - t) \cdots (\lambda_n - t)$$

where the λ_j are the *real* eigenvalues of S . It follows that $p(t)$ splits over $\mathbb{R}$.

By Schur's Lemma, there is an orthonormal basis β of V with respect to which $B := [T]_\beta$ is upper-triangular. However (Lemma 6.21), since T is self-adjoint and β is orthonormal,

$$B^* = [T^*]_\beta = [T]_\beta = B$$

Since B is simultaneously upper- and lower-triangular, it must be diagonal.

For the converse, suppose β is an orthonormal basis of eigenvectors of T . Then $[T]_\beta$ is diagonal with *real* entries. It follows that $[T^*]_\beta = [T]_\beta^* = [T]_\beta$, whence $T^* = T$ is self-adjoint. ■

The proof for the complex case is quite different. We can instantly apply the Fundamental Theorem of Algebra to invoke Schur and obtain a suitable basis. Since normality is a weaker condition than self-adjointness, the difficulty lies in showing that this basis consists entirely of eigenvectors.

Proof of Spectral Theorem for complex inner product spaces. Let $\dim V = n$ and T be normal. By Schur's Lemma, there exists an orthonormal basis $\beta = \{\mathbf{w}_1, \dots, \mathbf{w}_n\}$ for which $A := [T]_\beta$ is upper-triangular. We prove by induction that β is a basis of eigenvectors.

For the base case, $\mathbf{w}_1$ is an eigenvector by the upper-triangularity of A .

Now suppose $\mathbf{w}_1, \dots, \mathbf{w}_{k-1}$ are eigenvectors, with $T(\mathbf{w}_j) = \lambda_j \mathbf{w}_j$. By Lemma 6.34 $T^*(\mathbf{w}_j) = \overline{\lambda_j} \mathbf{w}_j$. By Corollary 6.19 and the orthonormality of β ,

$$j < k \implies A_{jk} = \langle T(\mathbf{w}_k), \mathbf{w}_j \rangle = \langle \mathbf{w}_k, T^*(\mathbf{w}_j) \rangle = \lambda_j \langle \mathbf{w}_k, \mathbf{w}_j \rangle = 0$$

However, since A is upper-triangular,

$$T(\mathbf{w}_k) = \sum_{j=1}^k A_{jk} \mathbf{w}_j = A_{kk} \mathbf{w}_k$$

whence $\mathbf{w}_k$ is an eigenvector of T . By induction, β is a basis of eigenvectors and A is diagonal.

Conversely, if β is an orthonormal basis of eigenvectors, then $[T]_\beta$ is diagonal and so $[T^*]_\beta = [T]_\beta^*$ is diagonal. Since diagonal matrices commute, we see that $T^*T = TT^*$: that is, T is normal. ■

Exercises. 6.4.1. For each linear operator T on an inner product space V , decide whether T is normal, self-adjoint, or neither. If the spectral theorem allows it, find an orthonormal basis of V consisting of eigenvectors of V .

(a) $V = \mathbb{R}^2$ with the dot product and $T \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 2a-b \\ -2a+5b \end{pmatrix}$

(b) $V = \mathbb{R}^3$ with the dot product and $T \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} -a-b \\ 5b \\ 4a-2b+5c \end{pmatrix}$

(c) $V = \mathbb{C}^2$ with the hermitian product and $T \begin{pmatrix} z \\ w \end{pmatrix} = \begin{pmatrix} 2z+iw \\ z+2iw \end{pmatrix}$

(d) $V = \mathbb{R}^4$ with $T : (\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3, \mathbf{e}_4) \mapsto (\mathbf{e}_3, \mathbf{e}_4, \mathbf{e}_1, \mathbf{e}_2)$

(e) $V = P_2(\mathbb{R})$ with $\langle f, g \rangle = \int_0^1 f(t)g(t) dt$ and $T(f) = f'$

(Hint: Don't compute T^* ! Instead assume T is normal and aim for a contradiction...)

6.4.2. Let $T(f)(x) = f'(x) + 4xf(0)$ where $T \in \mathcal{L}(P_1(\mathbb{R}))$ and $\langle f, g \rangle = \int_{-1}^1 f(t)g(t) dt$. Find an orthonormal basis of $P_1(\mathbb{R})$ with respect to which the matrix of T is upper-triangular.

6.4.3. Let S, T be operators on an inner product space V .

(a) Prove that $(ST)^* = T^*S^*$.

(Hint: Consider $\langle ST\mathbf{x}, \mathbf{y} \rangle \dots$)

(b) Suppose S, T are self-adjoint. Prove that ST is self-adjoint if and only if $ST = TS$.

6.4.4. Explain the difference between the converses in the two proofs of the Spectral Theorem: how do we come to different conclusions ($[T]_\beta$ self-adjoint versus normal)?

6.4.5. Let $T \in \mathcal{L}(V)$ where V is an inner product space, and let $W \leq V$ be T -invariant. Let $T|_W \in \mathcal{L}(W)$ be the restriction of T to W . Prove:

(a) T self-adjoint $\implies T|_W$ is self-adjoint.

(b) $W^\perp$ is T^* -invariant.

(c) If W is both T - and T^* -invariant, then $(T|_W)^* = (T^*)|_W$.

6.4.6. Let T be a normal operator on a finite-dimensional inner product space V . Prove that $\mathcal{N}(T^*) = \mathcal{N}(T)$ and that $\mathcal{R}(T^*) = \mathcal{R}(T)$.

(Hint: Use Lemma 6.34 and Exercise 6.3.4.)

6.4.7. Let T be self-adjoint on a finite-dimensional inner product space V . Prove that

$$\forall \mathbf{x} \in V, \quad \|T(\mathbf{x}) \pm i\mathbf{x}\|^2 = \|T(\mathbf{x})\|^2 + \|\mathbf{x}\|^2$$

Hence prove that $T - iI$ is invertible and that $[(T - iI)^{-1}]^* = (T + iI)^{-1}$.

6.4.8. (a) Suppose S is a normal operator on a finite-dimensional *complex* inner product space, all of whose eigenvalues are *real*. Prove that S is self-adjoint.

(b) Let T be a normal operator on a finite-dimensional *real* inner product space V whose characteristic polynomial splits. Prove that T is self-adjoint and that there exists an orthonormal basis of V of eigenvectors of T .

(Hint: read the proof of the spectral theorem and use part (a)...)

6.5 Unitary and Orthogonal Operators and Their Matrices

In this section we focus on length-preserving transformations of an inner product space.

Definition 6.38. A *linear isometry* of an inner product space V over $\mathbb{F}$ is a linear map T satisfying

$$\forall \mathbf{x} \in V, \|T(\mathbf{x})\| = \|\mathbf{x}\|$$

Example 6.39. Let $T = L_A \in \mathcal{L}(\mathbb{R}^2)$, where $A = \frac{1}{5} \begin{pmatrix} 4 & -3 \\ 3 & 4 \end{pmatrix}$. Then

$$\left\| T \begin{pmatrix} x \\ y \end{pmatrix} \right\|^2 = \left\| \frac{1}{5} \begin{pmatrix} 4x - 3y \\ 3x + 4y \end{pmatrix} \right\|^2 = \frac{1}{25} ((4x - 3y)^2 + (3x + 4y)^2) = x^2 + y^2 = \left\| \begin{pmatrix} x \\ y \end{pmatrix} \right\|^2$$

The matrix in this example is very special: its inverse is its transpose:

$$A^{-1} = \frac{1}{\frac{16}{25} + \frac{9}{25}} \begin{pmatrix} 4 & 3 \\ -3 & 4 \end{pmatrix} = \frac{1}{5} \begin{pmatrix} 4 & 3 \\ -3 & 4 \end{pmatrix} = A^T$$

It turns out that all linear isometries of a finite-dimensional inner product space have essentially this form. First we make a definition.

Definition 6.40. Let V be an inner product space. A *unitary operator* $T \in \mathcal{L}(V)$ is an invertible linear map satisfying $T^*T = I = TT^*$. If the base field is $\mathbb{R}$, we instead call these *orthogonal operators*.

The operator $T = L_A$ above is therefore orthogonal. We'll see shortly that the effect of the linear map is to rotate counter-clockwise around the origin in $\mathbb{R}^2$ by $\theta = \tan^{-1} \frac{3}{4}$.

Our main goal is the following:

Theorem 6.41. Let $T \in \mathcal{L}(V)$ where V is a finite-dimensional inner product space. Then T is a linear isometry if and only if it is a unitary/orthogonal operator.

Proof. Suppose T is a linear isometry. Since $(I - T^*T)^* = I^* - (T^*T)^* = I - T^*T$ is self-adjoint, by the spectral theorem, there exists an orthonormal basis of V of eigenvectors of $I - T^*T$. For any such $\mathbf{x}$ with eigenvalue λ ,

$$\begin{aligned} 0 &= \|\mathbf{x}\|^2 - \|T(\mathbf{x})\|^2 = \langle \mathbf{x}, \mathbf{x} \rangle - \langle T(\mathbf{x}), T(\mathbf{x}) \rangle = \langle \mathbf{x}, (I - T^*T)\mathbf{x} \rangle = \bar{\lambda} \|\mathbf{x}\|^2 \\ &\implies \lambda = 0 \end{aligned}$$

Since all eigenvalues are zero, $T^*T = I$. Since V is finite-dimensional, we also have $TT^* = I$ whence T is unitary.

Now suppose T is unitary. Then

$$\forall \mathbf{x}, \mathbf{y} \in V, \quad \langle \mathbf{x}, \mathbf{y} \rangle = \langle T^*T(\mathbf{x}), \mathbf{y} \rangle = \langle T(\mathbf{x}), T(\mathbf{y}) \rangle \quad (+)$$

In particular taking $\mathbf{x} = \mathbf{y}$ shows that T is an isometry. ■

The proof shows more:

Corollary 6.42. Every eigenvalue of a unitary operator has modulus 1.

Moreover, on a finite dimensional space, being unitary is equivalent to each of the following:

- (a) Preservation of the inner product $(\dagger)$. In particular, in a real inner product space isometries also preserve the angle θ between vectors since $\cos \theta = \frac{\langle \mathbf{x}, \mathbf{y} \rangle}{\|\mathbf{x}\| \|\mathbf{y}\|}$.
- (b) The existence of an orthonormal basis $\beta = \{\mathbf{w}_1, \dots, \mathbf{w}_n\}$ such that $T(\beta) = \{T(\mathbf{w}_1), \dots, T(\mathbf{w}_n)\}$ is also orthonormal.
- (c) That every orthonormal basis β of V must be mapped to an orthonormal basis $T(\beta)$.

The last first claim follows straight from $(\dagger)$, which holds even in infinite dimensions: if $T(\mathbf{w}) = \lambda \mathbf{w}$, then

$$\|\mathbf{w}\|^2 = \|T(\mathbf{w})\|^2 = \|\lambda \mathbf{w}\|^2 = |\lambda|^2 \|\mathbf{w}\|^2$$

Claims (b) and (c) are also worth proving explicitly, though they also quickly follow from $(\dagger)$: see Exercise 6.5.8..

Example 6.43. On $\mathbb{C}^2$ with the usual inner product, define $T = L_A$ where

$$A = \frac{i}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

The standard orthonormal basis $\epsilon = \{\mathbf{i}, \mathbf{j}\}$ is mapped to

$$T(\epsilon) = \left\{ \frac{i}{\sqrt{2}}(\mathbf{i} + \mathbf{j}), \frac{i}{\sqrt{2}}(\mathbf{j} - \mathbf{i}) \right\}$$

which is easily seen to be orthonormal. It follows that T is a unitary operator and that

$$T^* = T^{-1} = L_{A^{-1}}, \quad \text{where} \quad A^{-1} = -\frac{i}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

which is, of course, the matrix A^* . It is also easy to check that the characteristic polynomial is

$$p(t) = \det \begin{pmatrix} \frac{i}{\sqrt{2}} - t & \frac{i}{\sqrt{2}} \\ \frac{i}{\sqrt{2}} & -\frac{i}{\sqrt{2}} - t \end{pmatrix} = \frac{1}{2} + t^2 + \frac{1}{2} = t^2 + 1 \implies t = \pm i$$

whence the eigenvalues of T both have modulus 1.

Note that the discussion in the proof need not all hold for infinite dimensional operators, though some of the conclusions might.

Example 6.44. In the Fourier-series example, the function

$$T(f)(x) = e^{ix}f(x)$$

is linear, and moreover

$$\langle e^{ix}f(x), g(x) \rangle = \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{ix}f(x)\overline{g(x)} dx = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x)\overline{e^{-ix}g(x)} dx = \langle f(x), e^{-ix}g(x) \rangle$$

whence $T^*(f)(x) = e^{-ix}f(x)$. Indeed $T^* = T^{-1}$ and so T is a unitary operator. However T has no eigenvalues/eigenvectors:

$$T(f) = \lambda f \iff \forall x, e^{ix}f(x) = \lambda f(x)$$

which is impossible unless $f \equiv 0$. It can also be seen that the inner product space of continuous functions on $[-2\pi, 2\pi]$ with the above inner product has no orthonormal basis.^a

^aAn orthonormal set $\beta = \{f_k : k \in \mathbb{Z}\}$ can be found so that every function f equals an *infinite series* in the sense that $\|f - \sum a_k f_k\| = 0$, but these are not finite sums and so β is not a basis. Given that the norm is defined by an integral, this also isn't quite the same as saying that f equals $\sum a_k f_k$.

Unitary and Orthogonal Matrices

Suppose β is an orthonormal basis of V and that $T \in \mathcal{L}(V)$ is unitary. If $A = [T]_{\beta}$, then (Lemma 6.21)

$$A^*A = I = AA^*$$

It therefore makes sense to extend Definition 6.40 to matrices.

Definition 6.45. A *unitary matrix* is a matrix $A \in M_n(\mathbb{C})$ satisfying $A^*A = I$.
An *orthogonal matrix* satisfies $A^T A = I$.

Real unitary matrices are always referred to as orthogonal.⁴ Note that we don't need the condition $AA^* = I$ for matrices: $A^*A = I$ forces A to be invertible with $A^{-1} = A^*$... Both conditions are required only for unitary/orthogonal operators on *infinite-dimensional* spaces.

Before seeing some examples, we state a result which makes identifying these matrices very easy.

Corollary 6.46. A matrix $A \in M_n(\mathbb{R})$ is orthogonal if and only if its columns form an orthonormal basis of $\mathbb{R}^n$ with respect to the standard (dot) inner product.

A matrix $A \in M_n(\mathbb{C})$ is unitary if and only if its columns form an orthonormal basis of $\mathbb{C}^n$ with respect to the standard (hermitian) inner product.

The result follows immediately from Corollary 6.42: let $\beta = \{\mathbf{e}_1, \dots, \mathbf{e}_n\}$ be the standard (orthonormal) basis and $T = L_A$, then the columns of A are precisely the vectors $T(\mathbf{e}_j)$. We can also check this

⁴Complex orthogonal matrices exist (see Exercise 6.5.5.), though these do not have the same relationship to complex inner products as do unitary matrices.

directly by observing that

$$A = (\mathbf{w}_1 \cdots \mathbf{w}_n) \implies A^* A = \begin{pmatrix} \mathbf{w}_1^* \\ \vdots \\ \mathbf{w}_n^* \end{pmatrix} (\mathbf{w}_1 \cdots \mathbf{w}_n) \implies (A^* A)_{jk} = \mathbf{w}_j^* \mathbf{w}_k = \langle \mathbf{w}_k, \mathbf{w}_j \rangle$$

Clearly $A^* A = I$ if and only if $\{\mathbf{w}_1, \dots, \mathbf{w}_n\}$ is orthonormal.

Examples 6.47. 1. The matrix $A_\theta = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} \in M_2(\mathbb{R})$ is orthogonal for any θ . Example 6.39 is this with $\theta = \tan^{-1} \frac{3}{4}$. More generally (Exercise 6.5.6.), it can be seen that every real orthogonal 2×2 matrix has the form A_θ or

$$B_\theta = \begin{pmatrix} \cos \theta & \sin \theta \\ \sin \theta & -\cos \theta \end{pmatrix}$$

for some angle θ . The effect of the linear map L_{A_θ} is to rotate counter-clockwise by θ , while that of L_{B_θ} is to reflect across the line making angle $\frac{1}{2}\theta$ with the positive x -axis.

2. $A = \frac{1}{\sqrt{6}} \begin{pmatrix} \sqrt{2} & \sqrt{3} & 1 \\ \sqrt{2} & 0 & -2 \\ -\sqrt{2} & \sqrt{3} & -1 \end{pmatrix} \in M_3(\mathbb{R})$ is orthogonal.

3. The matrix $A = \frac{i}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$ in Example 6.43 is unitary.

Unitary and Orthogonal Equivalence

Suppose A is normal: $A^* A = A A^*$. It follows that A can be diagonalized with respect to an orthonormal basis $\beta = \{\mathbf{w}_1, \dots, \mathbf{w}_n\}$ of eigenvectors: $A \mathbf{w}_j = \lambda_j \mathbf{w}_j$. If we write $P = (\mathbf{w}_1 \cdots \mathbf{w}_n)$, then the columns of P are orthonormal and thus P is unitary! We can therefore write

$$A = P D P^{-1} = P \begin{pmatrix} \lambda_1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \lambda_n \end{pmatrix} P^*$$

Example 6.48. The matrix $A = \begin{pmatrix} 1+i & 1+i \\ -1-i & 1+i \end{pmatrix}$ is normal as can easily be checked. Its characteristic polynomial is

$$p(t) = t^2 - 2(1+i)t + 4i = (t-2i)(t-2)$$

with corresponding orthonormal eigenvectors

$$\mathbf{w}_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -i \end{pmatrix}, \quad \mathbf{w}_{2i} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ i \end{pmatrix}$$

We conclude that

$$A = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ -i & i \end{pmatrix} \begin{pmatrix} 2 & 0 \\ 0 & 2i \end{pmatrix} \left[\frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ -i & i \end{pmatrix} \right]^{-1} = \begin{pmatrix} 1 & 1 \\ -i & i \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix} \begin{pmatrix} 1 & i \\ 1 & -i \end{pmatrix}$$

This is an example of unitary equivalence:

Definition 6.49. Square matrices A, B are *unitarily equivalent* if there exists a unitary matrix P such that $B = P^*AP$. *Orthogonal equivalence* is similar: $B = P^TAP$.

The above discussion proves half the following:

Theorem 6.50. $A \in M_n(\mathbb{C})$ is normal if and only if it is unitarily equivalent to a diagonal matrix (the matrix of its eigenvalues).
Similarly, $A \in M_n(\mathbb{R})$ is symmetric if and only if it is orthogonally equivalent to a diagonal matrix.

Proof. We have almost finished the $(\Rightarrow)$ direction. if A is real and $A^T = A$, then it is self-adjoint. The spectral theorem says it has an orthonormal basis of eigenvectors.

For the converse, let D be diagonal, P unitary, and $A = P^*DP$. Then

$$A^*A = (P^*DP)^*P^*DP = P^*D^*PP^*DP = P^*\overline{D}DP = \dots = P^*DP(P^*DP)^* = AA^*$$

since $P^* = P^{-1}$ and because diagonal matrices commute: $\overline{D}D = D\overline{D}$.

In the special case where A is real and P is orthogonal, then A is symmetric:

$$A^T = (P^TDP)^T = P^TD^TP = P^TDP = A$$

Consider an $n \times n$ matrix A and take the linear map $T = L_A \in \mathcal{L}(\mathbb{C}^n)$. Applying Schur's Lemma, we see that there exists an orthonormal basis $\beta = \{\mathbf{w}_1, \dots, \mathbf{w}_n\}$ such that $B := [T]_\beta$ is upper-triangular. Also define

$$P = (\mathbf{w}_1 \cdots \mathbf{w}_n) = \begin{pmatrix} (\mathbf{w}_1)_1 & \cdots & (\mathbf{w}_n)_1 \\ \vdots & & \vdots \\ (\mathbf{w}_1)_n & \cdots & (\mathbf{w}_n)_n \end{pmatrix}$$

whose columns are the basis β and whose ik^{th} entry is $(\mathbf{w}_k)_i$. Since $[T(\mathbf{w}_j)]_\beta$ is the j^{th} column of B , we have

$$T(\mathbf{w}_j) = \sum_{k=1}^n \mathbf{w}_k B_{kj} \implies [T(\mathbf{w}_j)]_i = \sum_{k=1}^n (\mathbf{w}_k)_i B_{kj} = (PB)_{ij}$$

whence the matrix PB has j^{th} column $T(\mathbf{w}_j)$. However we also have

$$AP = (T(\mathbf{w}_1) \cdots T(\mathbf{w}_n)) \implies AP = PB \implies B = P^{-1}AP$$

Finally observe that P has orthonormal columns, so it is unitary: $P^{-1} = P^*$. If A were real, then we could have taken $T \in \mathcal{L}(\mathbb{R}^n)$, from which P would be orthogonal. The upshot is the following:

Corollary 6.51 (Schur's Lemma for Matrices). Every complex $n \times n$ matrix A is unitarily equivalent to an upper triangular matrix $B = P^*AP$.
If A is real, then it is orthogonally equivalent to a real upper triangular $B = P^TAP$.

Exercises. 6.5.1. For each matrix A find an orthogonal or unitary matrix P and a diagonal $D = P^*AP$.

$$(a) \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix} \quad (b) \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \quad (c) \begin{pmatrix} 2 & 3-3i \\ 3+3i & 5 \end{pmatrix} \quad (d) \begin{pmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{pmatrix}$$

6.5.2. Which of the following pairs of matrices are unitarily/orthogonally equivalent? Explain your answer.

$$(a) A = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \text{ and } B = \begin{pmatrix} 0 & 2 \\ 2 & 0 \end{pmatrix}$$

$$(b) A = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} \text{ and } B = \begin{pmatrix} 2 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$(c) A = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} \text{ and } B = \begin{pmatrix} 1 & 0 & 0 \\ 0 & i & 0 \\ 0 & 0 & -i \end{pmatrix}$$

6.5.3. Let $a, b \in \mathbb{C}$ be such that $|a|^2 + |b|^2 = 1$. Prove that every 2×2 matrix of the form $\begin{pmatrix} a & -e^{i\theta}\bar{b} \\ b & e^{i\theta}a \end{pmatrix}$ is unitary. Are these all the unitary 2×2 matrices? Prove or disprove...

6.5.4. If A, B are orthogonal, prove that AB and A^{-1} are also orthogonal. Do the same for unitary matrices.

(This proves that the orthogonal and unitary matrices form groups under matrix multiplication: usually denoted $O(n, \mathbb{R})$, $O(n, \mathbb{C})$ and $U(n)$.)

6.5.5. Check that $A = \frac{1}{3} \begin{pmatrix} 5 & -4i \\ 4i & 5 \end{pmatrix} \in M_2(\mathbb{C})$ is orthogonal. Is it unitary?

6.5.6. Supply the details of Exercise 6.47.1.

(Hints: $\beta = \{\mathbf{i}, \mathbf{j}\}$ is orthonormal, whence $\{A\mathbf{i}, A\mathbf{j}\}$ must be orthonormal. Now draw pictures to compute the result of rotating and reflecting the vectors $\mathbf{i}$ and $\mathbf{j}$.)

6.5.7. Prove that $A \in M_n(\mathbb{C})$ has an orthonormal basis of eigenvectors whose eigenvalues have modulus 1, if and only if A is unitary.

6.5.8. Prove the second and third parts of Corollary 6.42 for a finite-dimensional inner product space:

(a) If β is an orthonormal basis such that $T(\beta)$ is orthonormal, then T is unitary.

(b) If T is unitary, and η is an orthonormal basis, then $T(\eta)$ is an orthonormal basis.

6.5.9. Let T be a linear operator on a finite-dimensional inner product space V . If $\|T(\mathbf{x})\| = \|\mathbf{x}\|$ for all $\mathbf{x}$ in some orthonormal basis of V , must T be unitary? Prove or disprove.

6.5.10. Let T be a unitary operator on an inner product space V and let W be a finite-dimensional T -invariant subspace of V . Prove:

(a) $T(W) = W$ (Hint: show that $T|_W$ is injective...);

(b) $W^\perp$ is T -invariant.

6.5.11. Let W a subspace of an inner product space V such that $V = W \oplus W^\perp$. Define $T \in \mathcal{L}(V)$ by $T(\mathbf{u} + \mathbf{w}) = \mathbf{u} - \mathbf{w}$ where $\mathbf{u} \in W$ and $\mathbf{w} \in W^\perp$. Prove that T is unitary and self-adjoint.

6.6 Orthogonal Projections

We have already done most of the work for this in previous sections, though we start from a slightly different place.

Definition 6.52. Suppose $V = W_1 \oplus W_2$ is a direct sum decomposition of a vector space. The *projection onto W_1 along W_2* is the linear map $T \in \mathcal{L}(V)$ defined by

$$\forall \mathbf{r} \in W_1, \mathbf{n} \in W_2, \quad T(\mathbf{r} + \mathbf{n}) = \mathbf{r}$$

Clearly $W_1 = \mathcal{R}(T)$ and $W_2 = \mathcal{N}(T)$, hence our notation. A linear map is a *projection* precisely when subspaces W_1, W_2 with $V = W_1 \oplus W_2$ exist such that T has the above form.

Suppose T is a projection on an inner product space V : we say it is an *orthogonal projection* if

$$\mathcal{N}(T) = \mathcal{R}(T)^\perp \quad \text{and} \quad \mathcal{R}(T) = \mathcal{N}(T)^\perp$$

Given a subspace U satisfying $(U^\perp)^\perp = U$, the resulting projection with $\mathcal{R}(T) = U$ and $\mathcal{N}(T) = U^\perp$ is the *orthogonal projection onto U* : namely π_U .

Theorem 6.53. Let T be a linear operator on a vector space V . Then:

1. T is a projection if and only if $T^2 = T$.
2. When V is an inner product space, T is an orthogonal projection if and only if it has an adjoint and $T^2 = T = T^*$.

We prove the two parts separately.

Proof of part 1. Let $T \in \mathcal{L}(V)$ be given and throughout let $\mathbf{r} \in \mathcal{R}(T)$ and $\mathbf{n} \in \mathcal{N}(T)$.

If T is a projection, then $V = \mathcal{R}(T) \oplus \mathcal{N}(T)$ so that every vector in V has a unique representation $\mathbf{w} = \mathbf{r} + \mathbf{n}$. But then

$$T^2(\mathbf{r} + \mathbf{n}) = T(\mathbf{r}) = \mathbf{r} = T(\mathbf{r} + \mathbf{n}) \implies T^2 = T$$

Conversely, suppose $T^2 = T$. Our goal is to show that $V = \mathcal{R}(T) \oplus \mathcal{N}(T)$. For any $\mathbf{w} \in V$, write

$$\mathbf{w} = T(\mathbf{w}) + (\mathbf{w} - T(\mathbf{w})) \tag{*}$$

and observe:

- $T(\mathbf{w} - T(\mathbf{w})) = T(\mathbf{w}) - T^2(\mathbf{w}) = \mathbf{0} \implies \mathbf{w} - T(\mathbf{w}) \in \mathcal{N}(T)$;
- If $\mathbf{r} \in \mathcal{R}(T)$, then $\mathbf{r} = T(\mathbf{w})$ for some $\mathbf{w}$, whence

$$T(\mathbf{r}) = T^2(\mathbf{w}) = T(\mathbf{w}) = \mathbf{r} \tag{†}$$

In particular, $\mathcal{R}(T) \cap \mathcal{N}(T) = \{\mathbf{0}\}$ and (*) is therefore the unique decomposition of $\mathbf{w}$ into $\mathcal{R}(T)$ and $\mathcal{N}(T)$ parts. Moreover, (†) says that $T(\mathbf{r} + \mathbf{n}) = \mathbf{r}$, so that T is a projection. ■

Example 6.54. Let $T = L_A \in \mathcal{L}(\mathbb{R}^2)$, where $A = \frac{1}{5} \begin{pmatrix} 2 & 1 \\ 6 & 3 \end{pmatrix}$. Then T is a projection, since

$$\mathcal{R}(T) = \text{Span} \begin{pmatrix} 1 \\ 3 \end{pmatrix}, \quad \mathcal{N}(T) = \text{Span} \begin{pmatrix} 1 \\ -2 \end{pmatrix} \implies \mathbb{R}^2 = \mathcal{R}(T) \oplus \mathcal{N}(T)$$

and $T \begin{pmatrix} 1 \\ 3 \end{pmatrix} = \begin{pmatrix} 1 \\ 3 \end{pmatrix}$. Since $\mathcal{R}(T)$ and $\mathcal{N}(T)$ are not orthogonal spaces, the projection is not orthogonal. In fact, in $\mathbb{R}^2$, it is easy to classify all projections. There are three cases:

1. $T = I$ is the identity: $\mathcal{R}(T) = \mathbb{R}^2$ and $\mathcal{N}(T) = \{\mathbf{0}\}$;
2. $T = 0$ is the zero function: $\mathcal{R}(T) = \{\mathbf{0}\}$ and $\mathcal{N}(T) = \mathbb{R}^2$;
3. Choose two distinct 1-dimensional subspaces $\mathcal{R}(T) = \text{Span} \begin{pmatrix} a \\ b \end{pmatrix}$ and $\mathcal{N}(T) = \text{Span} \begin{pmatrix} c \\ d \end{pmatrix}$, then $T = L_A$ where

$$A = \frac{1}{ad - bc} \begin{pmatrix} a \\ b \end{pmatrix} (d - c) = \frac{1}{ad - bc} \begin{pmatrix} ad & -ac \\ bd & -bc \end{pmatrix}$$

Proof of part 2. Throughout, let $\mathbf{x} = \mathbf{x}_r + \mathbf{x}_n$ denote the decomposition of $\mathbf{x}$ according to $V = \mathcal{R}(T) \oplus \mathcal{N}(T)$.

($\Rightarrow$) Suppose T is an orthogonal projection. By assumption and by part 1, $\mathcal{R}(T)$ and $\mathcal{N}(T)$ are orthogonal subspaces, and $T^2 = T$. We have

$$\langle \mathbf{x}, T(\mathbf{y}) \rangle = \langle \mathbf{x}_r + \mathbf{x}_n, \mathbf{y}_r \rangle = \langle \mathbf{x}_r, \mathbf{y}_r + \mathbf{y}_n \rangle = \langle T(\mathbf{x}), \mathbf{y} \rangle$$

It follows that T is self-adjoint.

($\Leftarrow$) Suppose $T^2 = T = T^*$. By part 1, T is a projection and so $V = \mathcal{R}(T) \oplus \mathcal{N}(T)$. By the fundamental subspaces theorem (Exercise 6.3.4.),

$$\mathcal{R}(T) = \mathcal{R}(T^*) = \mathcal{N}(T)^\perp$$

Moreover, by Exercise 6.2.6., we also have $\mathcal{N}(T) \subseteq (\mathcal{N}(T)^\perp)^\perp = \mathcal{R}(T)^\perp$. Let $\mathbf{x} \in \mathcal{R}(T)^\perp$. Then

$$\forall \mathbf{w} \in V, 0 = \langle T(\mathbf{w}), \mathbf{x} \rangle = \langle \mathbf{w}, T(\mathbf{x}) \rangle \implies T(\mathbf{x}) = \mathbf{0} \implies \mathbf{x} \in \mathcal{N}(T)$$

We conclude that $\mathcal{N}(T) = \mathcal{R}(T)^\perp$, and that T is an orthogonal projection. ■

Example 6.55. By the above, it is clear that all orthogonal projections in $\mathbb{R}^2$ are either the identity, the zero map or are left-multiplication by a matrix of the form

$$A = \frac{1}{a^2 + b^2} \begin{pmatrix} a \\ b \end{pmatrix} (a \ b) = \frac{1}{a^2 + b^2} \begin{pmatrix} a^2 & ab \\ ab & b^2 \end{pmatrix}$$

where $\mathcal{R}(L_A) = \text{Span} \begin{pmatrix} a \\ b \end{pmatrix}$ and $\mathcal{N}(L_A) = \text{Span} \begin{pmatrix} -b \\ a \end{pmatrix}$. Otherwise said, choose a unit vector $\mathbf{w}$ and define $\pi_{\mathbf{w}} = \mathbf{w}\mathbf{w}^T$: this is, of course, precisely Definition 6.6.

Finite-dimensional Spaces and the Spectral Theorem

It should be clear that every projection T has (at most) two eigenspaces:

- $\mathcal{R}(T)$ is an eigenspace with eigenvalue 1;
- $\mathcal{N}(T)$ is an eigenspace with eigenvalue 0.

In a finite-dimensional space, if ρ is a basis of $\mathcal{R}(T)$ and η of $\mathcal{N}(T)$, then the matrix of T with respect to $\rho \cup \eta$ must have block form

$$[T]_{\rho \cup \eta} = \left(\begin{array}{c|c} I & 0 \\ \hline 0 & 0 \end{array} \right)$$

where $\text{rank } I = \text{rank } T$. In particular, every projection is diagonalizable.

For projections on a finite-dimensional vector space, the language of orthogonal projections allows us to rephrase Theorem 6.32.

Theorem 6.56 (Spectral Theorem, mk. II). *Let T be a normal/self-adjoint^a linear operator on a finite-dimensional inner product space V with distinct eigenvalues $\lambda_1, \dots, \lambda_k$ and corresponding eigenspaces $E_1, \dots, E_k$. Let $T_j \in \mathcal{L}(V)$ be the orthogonal projection onto E_j . Then:*

1. $V = E_1 \oplus \dots \oplus E_k$ is a direct sum of orthogonal subspaces, in particular, $E_j^\perp$ is the direct sum of the remaining eigenspaces;
2. $T_i T_j = 0$ if $i \neq j$;
3. $I_V = T_1 + \dots + T_k$;
4. $T = \lambda_1 T_1 + \dots + \lambda_k T_k$.

^aNormal if V is complex, self-adjoint if V is real.

Proof. 1. We know that T is diagonalizable and so V is the direct sum of the eigenspaces of T . By Lemma 6.34, the eigenvectors corresponding to distinct eigenvalues are orthogonal, whence the eigenspaces are mutually orthogonal. In particular, this says that

$$\hat{E}_j := \bigoplus_{i \neq j} E_i \leq E_j^\perp$$

Since V is finite-dimensional, Theorem 6.17 says $V = E_j \oplus E_j^\perp$, whence

$$\dim \hat{E}_j = \sum_{i \neq j} \dim E_i = \dim V - \dim E_j = \dim E_j^\perp \implies \hat{E}_j = E_j^\perp$$

2. This is clear by part 1, since $\mathcal{N}(T_j) = E_j^\perp = \hat{E}_j$.
3. Write $\mathbf{x} = \sum_{j=1}^k \mathbf{x}_j$ where each $\mathbf{x}_j \in E_j$. Then $T_j(\mathbf{x}) = \mathbf{x}_j$; now add...
4. $T(\mathbf{x}) = \sum_{j=1}^k T(\mathbf{x}_j) = \sum_{j=1}^k \lambda_j \mathbf{x}_j = \sum_{j=1}^k \lambda_j T_j(\mathbf{x})$. ■

Definition 6.57. The *spectrum* of a normal/self-adjoint operator T on a finite dimensional inner product space is its set of eigenvalues. The expressions in parts 3 and 4 are called, respectively, the *resolution of the identity* and the *spectral decomposition* of T .

Examples 6.58. 1. Recall Example 6.48 where we had a normal matrix $A = \begin{pmatrix} 1+i & 1+i \\ -1-i & 1+i \end{pmatrix}$ with orthonormal eigenvectors

$$\mathbf{w}_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -i \end{pmatrix}, \quad \mathbf{w}_{2i} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ i \end{pmatrix}$$

It is easy to see that the orthogonal projections are $T_2 = L_{B_2}$ and $T_{2i} = L_{B_{2i}}$, where

$$B_2 = \mathbf{w}_2 \mathbf{w}_2^* = \frac{1}{2} \begin{pmatrix} 1 \\ -i \end{pmatrix} (1 \ i) = \frac{1}{2} \begin{pmatrix} 1 & i \\ -i & 1 \end{pmatrix} \quad B_{2i} = \mathbf{w}_{2i} \mathbf{w}_{2i}^* = \frac{1}{2} \begin{pmatrix} 1 & -i \\ i & 1 \end{pmatrix}$$

Indeed observe that

$$B_2 + B_{2i} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \quad \text{and} \quad 2B_2 + 2iB_{2i} = \begin{pmatrix} 1 & i \\ -i & 1 \end{pmatrix} + \begin{pmatrix} i & 1 \\ 1 & i \end{pmatrix} = A$$

2. The matrix $A = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}$ has eigenvalues $2, -1, -1$ and an orthonormal eigenbasis

$$\{\mathbf{w}_1, \mathbf{w}_2, \mathbf{w}_3\} = \left\{ \frac{1}{\sqrt{3}} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}, \frac{1}{\sqrt{6}} \begin{pmatrix} 1 \\ 1 \\ -2 \end{pmatrix} \right\}$$

As eigenspaces, we have $E_2 = \text{Span}\{\mathbf{w}_1\}$ and $E_{-1} = \text{Span}\{\mathbf{w}_2, \mathbf{w}_3\}$. The orthogonal projections are then multiplication by the following matrices

$$B_2 = \mathbf{w}_1 \mathbf{w}_1^T = \frac{1}{3} \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}$$

$$B_{-1} = \mathbf{w}_2 \mathbf{w}_2^T + \mathbf{w}_3 \mathbf{w}_3^T = \frac{1}{2} \begin{pmatrix} 1 & -1 & 0 \\ -1 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix} + \frac{1}{6} \begin{pmatrix} 1 & 1 & -2 \\ 1 & 1 & -2 \\ -2 & -2 & 4 \end{pmatrix} = \frac{1}{3} \begin{pmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{pmatrix}$$

It is moreover easy to check the resolution of the identity and the spectral decomposition:

$$B_2 + B_{-1} = I \quad \text{and} \quad 2B_2 - B_{-1} = A$$

Exercises. 6.6.1. Compute the matrices of the orthogonal projections onto the following subspaces: in all cases we use the standard inner product.

- (a) $\text{Span} \left(\begin{pmatrix} 4 \\ -1 \end{pmatrix} \right)$ in $\mathbb{R}^2$
- (b) $\text{Span} \left\{ \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} \right\}$ in $\mathbb{R}^3$
- (c) $\text{Span} \left\{ \begin{pmatrix} i \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ i \\ 1 \end{pmatrix} \right\}$ in $\mathbb{C}^3$
- (d) $\text{Span} \left\{ \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} \right\}$ in $\mathbb{R}^3$ (watch out, these vectors aren't orthogonal!)

6.6.2. For each of the matrices in Exercise 6.5.1., compute the projections onto each eigenspace, verify the resolution of the identity and the spectral decomposition.

6.6.3. If W be a finite-dimensional subspace of an inner product space V . If $T = \pi_W$ is the orthogonal projection onto W , prove that then $I - T$ is the orthogonal projection onto $W^\perp$.

6.6.4. Let $T \in \mathcal{L}(V)$ where V is finite-dimensional.

- (a) If T is an orthogonal projection, prove that $\|T(\mathbf{x})\| \leq \|\mathbf{x}\|$ for all $\mathbf{x} \in V$.
- (b) Give an example of a projection for which the inequality in (a) is false.
- (c) If T is a projection for which $\|T(\mathbf{x})\| = \|\mathbf{x}\|$ for all $\mathbf{x} \in V$, what is T ?
- (d) If T is a projection for which $\|T(\mathbf{x})\| \leq \|\mathbf{x}\|$ for all $\mathbf{x} \in V$, prove that T is an orthogonal projection.

6.6.5. Let T be a normal operator on a finite-dimensional inner product space. If T is a projection, prove that it must be an orthogonal projection.

6.6.6. Let T be a normal operator on a finite-dimensional complex inner product space V . Use the spectral decomposition $T = \lambda_1 T_1 + \cdots + \lambda_k T_k$ to prove:

- (a) If T^n is the zero map for some $n \in \mathbb{N}$, then T is the zero map.
- (b) $U \in \mathcal{L}(V)$ commutes with T if and only if U commutes with each T_j .
- (c) There exists a normal $U \in \mathcal{L}(V)$ such that $U^2 = T$.
- (d) T is invertible if and only if $\lambda_j \neq 0$ for all j .
- (e) T is a projection if and only if every $\lambda_j = 0$ or 1 .
- (f) $T = -T^*$ if and only if every λ_j is imaginary.

6.7 The Singular Value Decomposition and the Pseudoinverse

Our first goal in this section is to extend the spectral theorem to any linear operator between vector spaces V, U .

Example 6.59. Let $T = L_A \in \mathcal{L}(\mathbb{R}^2, \mathbb{R}^3)$ where $A = \begin{pmatrix} 3 & 1 \\ 2 & -2 \\ 1 & 3 \end{pmatrix}$, then

$$T \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 4 \\ 0 \\ 4 \end{pmatrix} \quad \text{and} \quad T \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 2 \\ 4 \\ -2 \end{pmatrix}$$

If we take orthonormal bases

$$\beta = \{\mathbf{v}_1, \mathbf{v}_2\} = \left\{ \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix} \right\}, \quad \gamma = \{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\} = \left\{ \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \frac{1}{\sqrt{6}} \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}, \frac{1}{\sqrt{3}} \begin{pmatrix} 1 \\ -1 \\ -1 \end{pmatrix} \right\}$$

Then $T(\mathbf{v}_1) = 4\mathbf{u}_1$ and $T(\mathbf{v}_2) = 2\sqrt{3}\mathbf{u}_2$. With respect to these bases, we have

$$[T]_{\beta}^{\gamma} = \begin{pmatrix} 4 & 0 \\ 0 & 2\sqrt{3} \\ 0 & 0 \end{pmatrix}$$

which is as close to a diagonal matrix representation as we're going to be able to get.

In the example, the bases β, γ play a similar role to the orthonormal basis of eigenvectors diagonalizing a normal/self-adjoint operator as considered in the spectral theorem. The numbers 4 and $2\sqrt{3}$ behave very like eigenvalues. Our main result says that this can always be done.

Theorem 6.60 (Singular Value Decomposition). Suppose U, V are finite-dimensional inner product spaces and that $T \in \mathcal{L}(V, U)$ has rank r .

1. There exist orthonormal bases $\beta = \{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ of V and $\gamma = \{\mathbf{u}_1, \dots, \mathbf{u}_m\}$ of U and positive scalars $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_r$ such that

$$T(\mathbf{v}_j) = \begin{cases} \sigma_j \mathbf{u}_j & \text{if } j \leq r \\ \mathbf{0} & \text{otherwise} \end{cases}$$

2. Each $\mathbf{v}_j$ is an eigenvector of T^*T : indeed

$$T^*T(\mathbf{v}_j) = \begin{cases} \sigma_j^2 \mathbf{v}_j & \text{if } j \leq r \\ \mathbf{0} & \text{otherwise} \end{cases}$$

We conclude that the scalars σ_j are uniquely determined^a by T .

^aThe basis β is not, though it determines γ . The eigenspaces of T^*T are determined: these are, in order, spanned by the elements of β . Thus there is significant freedom in β , particularly if any of the σ_j 's are equal.

In the example, $\sigma_1 = 4, \sigma_2 = 2\sqrt{3}$ and $A^*A = \begin{pmatrix} 14 & 2 \\ 2 & 14 \end{pmatrix}$ which has eigenvalues $\sigma_1^2 = 16$ and $\sigma_2^2 = 12$.

Proof. 1. T^*T is self-adjoint: by the spectral theorem it has an orthonormal basis of eigenvectors $\beta = \{\mathbf{v}_1, \dots, \mathbf{v}_n\}$. Observe that

$$\langle T(\mathbf{v}_j), T(\mathbf{v}_k) \rangle = \langle T^*T(\mathbf{v}_j), \mathbf{v}_k \rangle = \lambda_j \delta_{jk}$$

Certainly every eigenvalue is a non-negative real number: $\lambda_j = \|T(\mathbf{v}_j)\|^2 \geq 0$. Exactly r of the eigenvalues are non-zero: by reordering the basis vectors if necessary, we may assume that

$$\lambda_1 \geq \dots \geq \lambda_r > 0$$

If $j \leq r$, define $\sigma_j := \sqrt{\lambda_j} > 0$ and $\mathbf{u}_j := \frac{1}{\sigma_j} T(\mathbf{v}_j)$, then the set $\{\mathbf{u}_1, \dots, \mathbf{u}_r\}$ is orthonormal. To finish, simply extend this to an orthonormal basis γ .

2. This is very easy given part 1:

$$\langle T^*(\mathbf{u}_j), \mathbf{v}_k \rangle = \langle \mathbf{u}_j, T(\mathbf{v}_k) \rangle = \langle \mathbf{u}_j, \sigma_k \mathbf{u}_k \rangle = \sigma_k \delta_{jk} = \langle \sigma_j \mathbf{v}_j, \mathbf{v}_k \rangle \implies T^*(\mathbf{u}_j) = \sigma_j \mathbf{v}_j$$

since β is a basis. We conclude that $T^*T(\mathbf{v}_j) = \sigma_j^2 \mathbf{v}_j$. ■

Definition 6.61. We call the numbers $\sigma_1, \dots, \sigma_r$ the *singular values* of T . If T is not maximum rank, we have additional singular values $\sigma_{r+1} = \dots = \sigma_{\min(m,n)} := 0$. If A is a matrix, then its singular values are the singular values of the linear map L_A .

It should be reasonably clear that if $V = U$ and if T is normal, then each $\mathbf{w}_j$ is an eigenvector of T and σ_j is the *modulus* of the corresponding eigenvalue (exercise).

Corollary 6.62 (Singular Value Decomposition for Matrices). Let $\sigma_1 \geq \dots \geq \sigma_r > 0$ be the non-zero singular values of a rank r matrix $A \in M_{m \times n}(\mathbb{F})$. Then there exist unitary matrices U, V such that

$$A = U\Sigma V, \quad \text{where} \quad \Sigma = \begin{pmatrix} \text{diag}(\sigma_1, \dots, \sigma_r) & O \\ O & O \end{pmatrix}$$

Proof. Let $T = L_A \in \mathcal{L}(\mathbb{F}^n, \mathbb{F}^m)$ and define $U = (\mathbf{u}_1, \dots, \mathbf{u}_m)$ and $V^* = (\mathbf{v}_1, \dots, \mathbf{v}_n)$ where the orthonormal basis vectors in the singular value theorem are the columns of the matrices. Clearly these are unitary matrices. Moreover, for each $\mathbf{v}_k$,

$$U\Sigma V\mathbf{v}_k = U\Sigma\mathbf{e}_k = \sigma_k U\mathbf{e}_k = \sigma_k \mathbf{u}_k = A\mathbf{v}_k$$

where $\mathbf{e}_k, \hat{\mathbf{e}}_k$ are the k^{th} standard basis vectors of $\mathbb{F}^m$ and $\mathbb{F}^n$ respectively. Since $U\Sigma V$ and A agree on a basis, they are equal. ■

Example 6.63. We find the singular value decomposition of the matrix $A = \begin{pmatrix} -1 & 7 \\ 3 & 4 \\ 7 & 1 \end{pmatrix}$.

According to the singular value theorem, we need to compute the orthonormal eigenpairs for

$$A^*A = \begin{pmatrix} -1 & 3 & 7 \\ 7 & 4 & 1 \end{pmatrix} \begin{pmatrix} -1 & 7 \\ 3 & 4 \\ 7 & 1 \end{pmatrix} = \begin{pmatrix} 59 & 12 \\ 12 & 66 \end{pmatrix}$$

When arranged in decreasing order, these are easily seen to be

$$(\mathbf{v}_1, \lambda_1) = \left(\frac{1}{5} \begin{pmatrix} 3 \\ 4 \end{pmatrix}, 75 \right) \quad \text{and} \quad (\mathbf{v}_2, \lambda_2) = \left(\frac{1}{5} \begin{pmatrix} -4 \\ 3 \end{pmatrix}, 50 \right)$$

It follows that the singular values are $\sigma_1 = \sqrt{75} = 5\sqrt{3}$ and $\sigma_2 = 5\sqrt{2}$. Now compute the other basis vectors:

$$\mathbf{u}_1 := \frac{1}{\sigma_1} A \mathbf{v}_1 = \frac{1}{25\sqrt{3}} \begin{pmatrix} 25 \\ 25 \\ 25 \end{pmatrix} = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \quad \mathbf{u}_2 := \frac{1}{\sigma_2} A \mathbf{v}_2 = \frac{1}{25\sqrt{2}} \begin{pmatrix} 25 \\ 0 \\ -25 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}$$

and we take $\mathbf{u}_3 = \frac{1}{\sqrt{6}} \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix}$ to complete the orthonormal basis $\gamma = \{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$. We conclude that

$$A = U\Sigma V = \begin{pmatrix} \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & 0 & \frac{-2}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & \frac{-1}{\sqrt{2}} & \frac{1}{\sqrt{6}} \end{pmatrix} \begin{pmatrix} 5\sqrt{3} & 0 \\ 0 & 5\sqrt{2} \\ 0 & 0 \end{pmatrix} \begin{pmatrix} \frac{3}{5} & \frac{-4}{5} \\ \frac{4}{5} & \frac{3}{5} \end{pmatrix}^T$$

It is typically much harder to find singular values in non-standard inner product spaces since computation of the adjoint is typically so difficult. Here is a classic example.

Example 6.64. On the inner product spaces $P_2(\mathbb{R})$ and $P_1(\mathbb{R})$ with $\langle f, g \rangle = \int_0^1 f(x)g(x) dx$, we have orthonormal bases

$$\beta = \{\sqrt{5}(6x^2 - 6x + 1), \sqrt{3}(2x - 1), 1\}, \quad \gamma = \{\sqrt{3}(2x - 1), 1\}$$

Now let $T = \frac{d}{dx}$. Since the bases are orthonormal, it is easy to find the matrices of T and its adjoint, etc:

$$[T]_{\beta}^{\gamma} = A = \begin{pmatrix} 2\sqrt{15} & 0 & 0 \\ 0 & 2\sqrt{3} & 0 \end{pmatrix}, \quad [T^*]_{\gamma}^{\beta} = A^* = \begin{pmatrix} 2\sqrt{15} & 0 \\ 0 & 2\sqrt{3} \\ 0 & 0 \end{pmatrix} \implies A^*A = \begin{pmatrix} 60 & 0 & 0 \\ 0 & 12 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

Since A^*A is diagonal, it follows that the orthonormal bases β, γ are already ordered in line with the singular values $\sigma_1 = 2\sqrt{15}, \sigma_2 = 2\sqrt{3}$.

Since β, γ are already the correct orthonormal bases for the singular value theorem, the matrix singular value decomposition with respect to these is completely uninteresting: $A = IAI$!

The Pseudoinverse

Given the singular values, it is straightforward to define something that lookslike the inverse of a linear transformation, but which makes sense even when the transformation is not invertible!

Definition 6.65. Suppose $T \in \mathcal{L}(V, U)$ is a linear map between finite-dimensional inner product spaces and that we've found its singular value decomposition. The *pseudoinverse* of T is the linear map $T^\dagger \in \mathcal{L}(U, V)$ defined by

$$T^\dagger(\mathbf{u}_j) = \begin{cases} \frac{1}{\sigma_j} \mathbf{v}_j & \text{if } j \leq r \\ \mathbf{0} & \text{otherwise} \end{cases}$$

If A is a matrix, then its pseudoinverse $A^\dagger$ is the matrix of $L_A^\dagger$ with respect to the standard bases.

Restricted to $\mathcal{N}(T)^\perp = \text{Span}\{\mathbf{v}_1, \dots, \mathbf{v}_r\}$ and $\mathcal{R}(T) = \text{Span}\{\mathbf{u}_1, \dots, \mathbf{u}_r\}$ we obtain an isomorphism

$$T|_{\mathcal{N}(T)^\perp} : \mathcal{N}(T)^\perp \rightarrow \mathcal{R}(T) \quad \text{with inverse} \quad T^\dagger|_{\mathcal{R}(T)}$$

Example 6.66. If $T = L_A$ where $A = \begin{pmatrix} -1 & 7 \\ 3 & 4 \\ 7 & 1 \end{pmatrix}$, then we have pseudoinverse

$$T^\dagger := \frac{1}{\sigma_1} \mathbf{v}_1 \mathbf{u}_1^* + \frac{1}{\sigma_2} \mathbf{v}_2 \mathbf{u}_2^*$$

which has matrix

$$A^\dagger = \frac{1}{5\sqrt{3} \cdot 5 \cdot \sqrt{3}} \begin{pmatrix} 3 \\ 4 \end{pmatrix} (1 \ 1 \ 1) + \frac{1}{5\sqrt{2} \cdot 5 \cdot \sqrt{2}} \begin{pmatrix} -4 \\ 3 \end{pmatrix} (1 \ 0 \ -1) = \frac{1}{150} \begin{pmatrix} -6 & 6 & 18 \\ 17 & 8 & -1 \end{pmatrix}$$

It is easy to check that

$$A^\dagger A = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad \text{and} \quad AA^\dagger = \frac{1}{6} \begin{pmatrix} 5 & 2 & -1 \\ 2 & 2 & 2 \\ -1 & 2 & 5 \end{pmatrix}$$

The first result is unsurprising: since T has rank 2 the restriction $T : \mathbb{R}^2 \rightarrow \mathcal{R}(T)$ is a isomorphism! Finally, observe that $TT^\dagger = \mathbf{u}_1 \mathbf{u}_1^* + \mathbf{u}_2 \mathbf{u}_2^* = \pi_{\mathbf{u}_1} + \pi_{\mathbf{u}_2}$ is precisely the *orthogonal projection* onto $\mathcal{R}(T) = \text{Span}\{\mathbf{u}_1, \mathbf{u}_2\}$: indeed, given A as above, it is easy to check that

$$A(A^T A)^{-1} A^T = AA^\dagger$$

is the projection matrix onto the column space of A , as discussed in Theorem 6.27.

In fact this link-up with projections happens in general:

Corollary 6.67. $TT^\dagger = \pi_{\mathcal{R}(T)}$ and $T^\dagger T = \pi_{\mathcal{N}(T)^\perp}$

Finally we check that the pseudoinverse of the derivative operator behaves roughly as expected.

Example 6.68. The pseudoinverse of $T = \frac{d}{dx} : P_2(\mathbb{R}) \rightarrow P_1(\mathbb{R})$, as seen in Example 6.64, maps

$$\begin{aligned} T^\dagger(\sqrt{3}(2x-1)) &= \frac{1}{2\sqrt{15}}\sqrt{5}(6x^2-6x+1) = \frac{1}{2\sqrt{3}}(6x^2-6x+1) \\ T^\dagger(1) &= \frac{1}{2\sqrt{3}}\sqrt{3}(2x-1) = x - \frac{1}{2} \\ \implies T^\dagger(a+bx) &= T^\dagger\left(a + \frac{b}{2} + \frac{b}{2\sqrt{3}}\sqrt{3}(2x-1)\right) = \left(a + \frac{b}{2}\right)\left(x - \frac{1}{2}\right) + \frac{b}{12}(6x^2-6x+1) \\ &= \frac{b}{2}x^2 + ax - \frac{b}{6} \end{aligned}$$

As expected, the pseudoinverse of ‘differentiation’ returns a particular choice of anti-derivative.

Exercises. 6.7.1. Find the ingredients β, γ and the singular values for each of the following:

(a) $T \in \mathcal{L}(\mathbb{R}^2, \mathbb{R}^3)$ where $T\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x+y \\ x-y \end{pmatrix}$

(b) $T : P_2(\mathbb{R}) \rightarrow P_1(\mathbb{R})$ and $T(f) = f''$ where $\langle f, g \rangle := \int_0^1 f(x)g(x) dx$

(c) $V = W = \text{Span}\{1, \sin x, \cos x\}$ and $\langle f, g \rangle = \int_0^{2\pi} f(x)g(x) dx$, with $T(f) = f' + 2f$

6.7.2. Find a singular value decomposition of each of the matrices:

(a) $\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$ (b) $\begin{pmatrix} 1 & 0 & 1 \\ 1 & 0 & -1 \end{pmatrix}$ (c) $\begin{pmatrix} 1 & 1 & 1 \\ 1 & -1 & 0 \\ 1 & 0 & -1 \end{pmatrix}$

6.7.3. Find an explicit formula for $T^\dagger$ in each of the examples in Exercise 6.7.1..

6.7.4. Find the pseudoinverse of each of the matrices in Exercise 6.7.2..

6.7.5. Suppose $T : V \rightarrow U$ is written according to the singular value theorem. Prove that γ is a basis of eigenvectors of TT^* with the *same* non-zero eigenvalues as T^*T , including repetitions.

6.7.6. Suppose $T = \mathcal{L}(V)$ is a normal operator. Prove that each $\mathbf{v}_j$ in the singular value theorem may be chosen to be an eigenvector of T and that σ_j is the *modulus* of the corresponding eigenvalue.

6.7.7. Let V, U be finite-dimensional inner product spaces and $T \in \mathcal{L}(V, U)$. Prove:

(a) If T is injective, then T^*T is invertible and $T^\dagger = (T^*T)^{-1}T^*$.

(b) If T is surjective, then TT^* is invertible and $T^\dagger = T^*(TT^*)^{-1}$.

6.7.8. Suppose $A \in M_{m \times n}(\mathbb{R})$ and $\mathbf{b} \in \mathbb{R}^m$ where $m > n$, so that the system of equations $A\mathbf{x} = \mathbf{b}$ is overdetermined. Let $\mathbf{z} = A^\dagger \mathbf{b}$.

(a) If $A\mathbf{x} = \mathbf{b}$ is consistent (has a solution), prove that $\mathbf{z}$ is the solution with minimal norm.

(b) If $A\mathbf{x} = \mathbf{b}$ is inconsistent, prove that $\mathbf{z}$ is the unique minimizer of $\|A\mathbf{z} - \mathbf{b}\|$ with minimal norm.

6.7.9. For the following systems, find the minimal norm solution to the first and the vector which comes closest to solving the second:

$$\begin{cases} -x + 7y = 6 \\ 3x + 4y = 7 \\ 7x + y = 8 \end{cases} \quad \begin{cases} -x + 7y = 1 \\ 3x + 4y = 0 \\ 7x + y = 0 \end{cases}$$

6.8 Bilinear and Quadratic Forms

In this section we slightly generalize the idea of an inner products. Throughout, V is a vector space over a field $\mathbb{F}$: unlike the rest of this chapter, V need not be an inner product space and $\mathbb{F}$ can be any field (not just $\mathbb{R}$ or $\mathbb{C}$).

Definition 6.69. A bilinear form $B : V \times V \rightarrow \mathbb{F}$ is a function which is linear in each entry when the other is held fixed. That is: $\forall \mathbf{v}, \mathbf{x}, \mathbf{y} \in V, \lambda \in \mathbb{R}$,

$$\langle \lambda \mathbf{x} + \mathbf{y}, \mathbf{v} \rangle = \lambda \langle \mathbf{x}, \mathbf{v} \rangle + \langle \mathbf{y}, \mathbf{v} \rangle, \quad \langle \mathbf{v}, \lambda \mathbf{x} + \mathbf{y} \rangle = \lambda \langle \mathbf{v}, \mathbf{x} \rangle + \langle \mathbf{v}, \mathbf{y} \rangle$$

Additionally, B is *symmetric* if $\forall \mathbf{x}, \mathbf{y} \in V, \langle \mathbf{x}, \mathbf{y} \rangle = \langle \mathbf{y}, \mathbf{x} \rangle$.

Examples 6.70. 1. If V is a *real* inner product space, then the inner product $\langle , \rangle$ is a symmetric bilinear form. Note that a *complex* inner product is *not bilinear*!

2. If $A \in M_n(\mathbb{F})$, then $B(\mathbf{v}, \mathbf{w}) := \mathbf{v}^T A \mathbf{w}$ is a bilinear form on $\mathbb{F}^n$. For instance, on $\mathbb{R}^2$, let

$$B(\mathbf{v}, \mathbf{w}) = \mathbf{v}^T \begin{pmatrix} 1 & 2 \\ 2 & 0 \end{pmatrix} \mathbf{w} = v_1 w_1 + 2v_1 w_2 + 2v_2 w_1$$

This defines a symmetric bilinear form. Note that B *isn't* an inner product on $\mathbb{R}^2$ since it isn't positive definite: for example,

$$B(\mathbf{j}, \mathbf{j}) = 0$$

3. On the vector space $V = \mathbb{Z}_3^2$ over the finite field $\mathbb{Z}_3 = \{0, 1, 2\}$ of remainders modulo 3, define

$$B(\mathbf{v}, \mathbf{w}) = \mathbf{v}^T \begin{pmatrix} 2 & 0 \\ 1 & 1 \end{pmatrix} \mathbf{w} = 2v_1 w_1 + v_2 w_1 + v_2 w_2$$

For instance, $B\left(\begin{pmatrix} 2 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 2 \end{pmatrix}\right) = 1$. This clearly isn't symmetric, since $B\left(\begin{pmatrix} 1 \\ 2 \end{pmatrix}, \begin{pmatrix} 2 \\ 0 \end{pmatrix}\right) = 2$.

As seen in the examples, we often make use of a matrix.

Definition 6.71. Let B be a bilinear form on a finite-dimensional vector space V with basis $\beta = \{\mathbf{v}_1, \dots, \mathbf{v}_n\}$. The matrix $A \in M_n(\mathbb{F})$ with ij^{th} entry

$$A_{ij} = B(\mathbf{v}_i, \mathbf{v}_j)$$

is the *matrix of B with respect to β* .

The correspondence should be clear: given $\mathbf{x}, \mathbf{y} \in V$ we may compute their co-ordinate vectors $[\mathbf{x}]_\beta, [\mathbf{y}]_\beta$ with respect to β , then

$$B(\mathbf{x}, \mathbf{y}) = [\mathbf{x}]_\beta^T A [\mathbf{y}]_\beta$$

In particular, given a basis β , the set of bilinear forms on V is in bijective correspondence with the set

$M_n(\mathbb{F})$. Moreover, since the transpose of a scalar is the same scalar,

$$B(\mathbf{y}, \mathbf{x}) = \left([\mathbf{y}]_{\beta}^T A [\mathbf{x}]_{\beta} \right)^T = [\mathbf{x}]_{\beta}^T A^T [\mathbf{y}]_{\beta}$$

Evaluating this on all pairs of basis elements, we see that

$$\begin{aligned} B \text{ symmetric} &\implies \forall i, j, B(\mathbf{v}_i, \mathbf{v}_j) = B(\mathbf{v}_j, \mathbf{v}_i) \\ &\implies \forall i, j, A_{ij} = A_{ji} \\ &\implies A^T = A \text{ (} A \text{ symmetric)} \\ &\implies B \text{ symmetric} \end{aligned}$$

We have therefore proved:

Lemma 6.72. *A bilinear form on a finite-dimensional vector space is symmetric if and only if its matrix with respect to any (and all) bases is symmetric.*

Given the relationship of the co-ordinate vectors with the matrix of a bilinear form, it is easy to see that if γ is another basis of V , then

$$B(\mathbf{x}, \mathbf{y}) = [\mathbf{x}]_{\beta}^T A [\mathbf{y}]_{\beta} = (Q_{\gamma}^{\beta} [\mathbf{x}]_{\gamma})^T A (Q_{\gamma}^{\beta} [\mathbf{y}]_{\gamma}) = [\mathbf{x}]_{\gamma}^T \left((Q_{\gamma}^{\beta})^T A Q_{\gamma}^{\beta} \right) [\mathbf{y}]_{\gamma}$$

where $Q_{\beta}^{\gamma} = [I]_{\beta}^{\gamma}$ is the change or co-ordinate matrix from β to γ . In particular,

Lemma 6.73. *If A is the matrix of B with respect to some basis, then every other matrix of B has the form $Q^T A Q$ for some invertible Q .*

Example 6.74. Recall Example 6.70.2, where $A = \begin{pmatrix} 1 & 2 \\ 2 & 0 \end{pmatrix}$ is the matrix of B with respect to the standard basis $\beta = \{\mathbf{i}, \mathbf{j}\}$ of $\mathbb{R}^2$. Let $\gamma = \{\mathbf{w}_1, \mathbf{w}_2\} := \{\mathbf{i}, \mathbf{j} - 2\mathbf{i}\}$, then

$$\begin{aligned} B(\mathbf{w}_1, \mathbf{w}_1) &= 1, \quad B(\mathbf{w}_2, \mathbf{w}_1) = B(\mathbf{w}_1, \mathbf{w}_2) = B(\mathbf{i}, \mathbf{j}) - 2B(\mathbf{i}, \mathbf{i}) = 0, \\ B(\mathbf{w}_2, \mathbf{w}_2) &= (-2 \ 1) \begin{pmatrix} 1 & 2 \\ 2 & 0 \end{pmatrix} \begin{pmatrix} -2 \\ 1 \end{pmatrix} = -4 \end{aligned}$$

It follows that the matrix of B with respect to γ is

$$\begin{pmatrix} 1 & 0 \\ 0 & -4 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ -2 & 1 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 2 & 0 \end{pmatrix} \begin{pmatrix} 1 & -2 \\ 0 & 1 \end{pmatrix} = Q^T A Q$$

where $Q = Q_{\beta}^{\gamma} = ([\mathbf{w}_1]_{\beta} \ [\mathbf{w}_2]_{\beta})$ is plainly the change of co-ordinate matrix from γ to β .

This example gives us a clue regarding the *diagonalization* of a bilinear form: the change of co-ordinate matrix in this case is an *elementary matrix* $E_{12}^{(-2)}$. Recall:

- As a row operation (left-multiplication), $E_{21}^{(-2)} = Q^T$ adds to the second row -2 times the first.
- As a column operation (right-multiplication), $E_{12}^{(-2)} = Q$ adds to the second column -2 times the first.

By repeatedly applying simultaneous row and column operations, we can diagonalize almost any symmetric bilinear form (caveat below...).

Example 6.75. Diagonalize $B(\mathbf{v}, \mathbf{w}) = \mathbf{v}^T A \mathbf{w} = \mathbf{v}^T \begin{pmatrix} 1 & -2 & 3 \\ -2 & 0 & 4 \\ 3 & 4 & -1 \end{pmatrix} \mathbf{w}$

We perform a sequence of row and column operations to clean out all non-diagonal entries. We only need elementary matrices of type III, which we'll denote $E_{ij}^{(\lambda)}$: this is the identity matrix with an additional λ in the ij^{th} entry.

$$E_{21}^{(2)} A E_{12}^{(2)} = \begin{pmatrix} 1 & 0 & 3 \\ 0 & -4 & 10 \\ 3 & 10 & -1 \end{pmatrix} \quad (\text{add twice the first row to the second, columns similarly})$$

$$E_{31}^{(-3)} E_{21}^{(2)} A E_{12}^{(2)} E_{13}^{(-3)} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -4 & 10 \\ 0 & 10 & -10 \end{pmatrix} \quad (\text{subtract 3 times the first row from the third, etc.})$$

$$E_{23}^{(1)} E_{31}^{(-3)} E_{21}^{(2)} A E_{12}^{(2)} E_{13}^{(-3)} E_{32}^{(1)} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & -10 \end{pmatrix} \quad (\text{add the third row to the second, etc.})$$

If $\beta = \{\mathbf{i}, \mathbf{j}, \mathbf{k}\}$ is the standard basis, we therefore have change of co-ordinate matrix

$$Q = Q_\gamma^\beta = E_{12}^{(2)} E_{13}^{(-3)} E_{32}^{(1)} = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & -3 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix} = \begin{pmatrix} 1 & -1 & -3 \\ 0 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix}$$

With respect to the basis $\gamma = \left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} -1 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} -3 \\ 0 \\ 1 \end{pmatrix} \right\}$ the symmetric bilinear form B is diagonal.

Theorem 6.76. *If B is diagonalizable on a finite-dimensional vector space, then it is symmetric. Conversely, if B is symmetric and the field $\mathbb{F}$ does not have characteristic 2, then B is diagonalizable.*

Example 6.77. Why the problem with characteristic 2? Consider $B = \mathbf{v}^T \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \mathbf{w}$. If we apply our algorithm:

$$E_{12}^{(1)} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} E_{21}^{(1)} = \begin{pmatrix} 2 & 1 \\ 1 & 0 \end{pmatrix} \quad (\text{add the second row to the first, columns similarly})$$

$$E_{21}^{(-1/2)} E_{12}^{(1)} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} E_{21}^{(1)} E_{12}^{(-1/2)} = \begin{pmatrix} 2 & 0 \\ 0 & -\frac{1}{2} \end{pmatrix} \quad (\text{subtract } \frac{1}{2} \text{ row 1 from row 2, etc.})$$

Since $Q = E_{21}^{(1)} E_{12}^{(-1/2)} = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & -\frac{1}{2} \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & -\frac{1}{2} \\ 1 & \frac{1}{2} \end{pmatrix}$ we see that B is diagonal with respect to the basis $\gamma = \left\{ \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \begin{pmatrix} -1/2 \\ 1/2 \end{pmatrix} \right\}$. If the field has characteristic 2, however, then $\frac{1}{2}$ is meaningless! Perhaps surprisingly, the matrix of B is the same with respect to *any* basis of $\mathbb{Z}_2^2$: B is therefore non-diagonalizable!

Proof. The first part is easy: B is diagonalizable if there exists a basis with respect to which its matrix A is diagonal. Since A is symmetric, so is B .

The converse is more difficult,^a. Suppose B is non-zero, for otherwise this is trivial. First observe that there exists a vector $\mathbf{x} \in V$ such that $B(\mathbf{x}, \mathbf{x}) \neq 0$: choose any two vectors $\mathbf{u}, \mathbf{v} \in V$ such that $B(\mathbf{u}, \mathbf{v}) \neq 0$ and if both $B(\mathbf{u}, \mathbf{u}) = 0 = B(\mathbf{v}, \mathbf{v})$, then $\mathbf{x} = \mathbf{u} + \mathbf{v}$ satisfies

$$B(\mathbf{x}, \mathbf{x}) = 2B(\mathbf{u}, \mathbf{v}) \neq 0$$

This is where the characteristic $\neq 2$ condition comes in!

Now proceed by induction on the dimension n of V . If $n = 1$ the result is trivial: every bilinear form on V is symmetric since it has the form $B(x, y) = axy$ for some $a \in \mathbb{F}$.

Now fix $n \in \mathbb{N}$ and assume the result holds for all non-zero symmetric bilinear forms on all vector spaces with dimension $\leq n$. Let $\dim V = n + 1$. By the above, $\exists \mathbf{x} \in V$ such that $B(\mathbf{x}, \mathbf{x}) \neq 0$. Consider the linear map

$$T : V \rightarrow \mathbb{F} : \mathbf{v} \mapsto B(\mathbf{x}, \mathbf{v})$$

Since $\text{rank } T = 1$, we see that $\text{null } T = n$ and so $N(T) \leq V$ is a dimension n subspace. Clearly B is symmetric when restricted to $N(T)$ and thus, by the induction hypothesis, is diagonalizable. Choose a basis β of $N(T)$ such that the matrix of $B|_{N(T)}$ with respect to β is diagonal. Then B is clearly diagonal with respect to the basis $\beta \cup \{\mathbf{x}\}$. ■

^aIf the field is $\mathbb{R}$ it is easy: $A^T = A$ is symmetric and thus orthogonally equivalent to a diagonal matrix...

Corollary 6.78. Over any field $\mathbb{F}$ of characteristic $\neq 2$, every symmetric matrix is similar to a diagonal matrix.

Quadratic Forms

Definition 6.79. To every symmetric bilinear form $B : V \times V \rightarrow \mathbb{F}$ is associated a *quadratic form*

$$K : V \rightarrow \mathbb{F} : \mathbf{v} \mapsto B(\mathbf{v}, \mathbf{v})$$

Indeed a function K is a quadratic form precisely when such a symmetric bilinear form exists.

Examples 6.80. 1. If B is a real inner product on V , then $K(\mathbf{v}) = \langle \mathbf{v}, \mathbf{v} \rangle = \|\mathbf{v}\|^2$ is the square of the norm. This partly explains the term 'quadratic'.

2. Let $\dim V = n$ and A be the matrix of B with respect to a basis β . By the symmetry of A ,

$$K(\mathbf{x}) = \mathbf{x}^T A \mathbf{x} = \sum_{i,j=1}^n x_i A_{ij} x_j = \sum_{1 \leq i \leq j \leq n} \tilde{a}_{ij} x_i x_j \quad \text{where } \tilde{a}_{ij} = \begin{cases} A_{ij} & \text{if } i = j \\ 2A_{ij} & \text{if } i \neq j \end{cases}$$

E.g., $K(\mathbf{x}) = 3x_1^2 + 4x_2^2 - 2x_1x_2$ corresponds to the bilinear form $B(\mathbf{x}, \mathbf{y}) = \mathbf{x}^T \begin{pmatrix} 3 & -1 \\ -1 & 4 \end{pmatrix} \mathbf{y}$

An application of quadratic forms and their relationship to symmetric (diagonalizable) bilinear forms is to the study of quadratic manifolds: these are conics in $\mathbb{R}^2$ and quadratic surfaces in $\mathbb{R}^3$. For example, the general non-zero quadratic form on $\mathbb{R}^2$ is

$$K(\mathbf{x}) = ax^2 + 2bxy + cy^2 \longleftrightarrow B(\mathbf{v}, \mathbf{w}) = \mathbf{v}^T \begin{pmatrix} a & b \\ b & c \end{pmatrix} \mathbf{w}$$

Diagonalizing with respect to an orthonormal basis $\{\mathbf{v}_1, \mathbf{v}_2\}$, there exist scalars λ_1, λ_2 with

$$K(t_1\mathbf{v}_1 + t_2\mathbf{v}_2) = \lambda_1 t_1^2 + \lambda_2 t_2^2$$

With respect to this basis the general conic has the form

$$\lambda_1 t_1^2 + \lambda_2 t_2^2 + \mu_1 t_1 + \mu_2 t_2 = \eta, \quad \lambda_1, \lambda_2, \mu_1, \mu_2, \eta \in \mathbb{R}$$

If the λ_i are non-zero, we may complete the square via the linear transformations $s_j = t_j + \frac{\mu_j}{2\lambda_j}$. The canonical forms are then recovered:

Parabola λ_1 or $\lambda_2 = 0$.

Ellipse $\lambda_1 s_1^2 + \lambda_2 s_2^2 = k$ where λ_1, λ_2 have the same sign.

Hyperbola $\lambda_1 s_1^2 + \lambda_2 s_2^2 = k$ where λ_1, λ_2 have opposite signs.

Since we only applied a rotation (change to the orthonormal basis $\{\mathbf{v}_1, \mathbf{v}_2\}$) and translation (completing the square), it is clear that every conic may be recovered thus from the canonical forms.

Example 6.81. We describe and plot the conic with equation $7x^2 + 24xy = 144$.

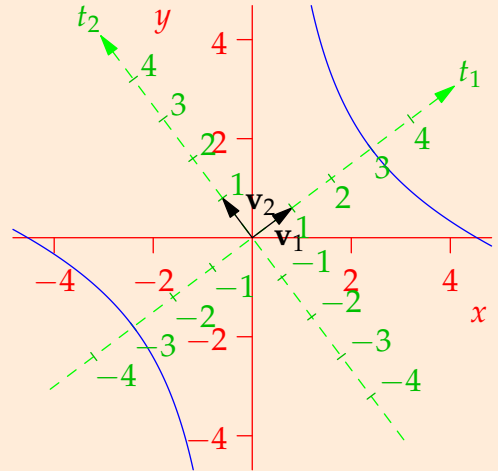
The matrix of the associated bilinear form is $\begin{pmatrix} 7 & 12 \\ 12 & 0 \end{pmatrix}$ which has orthonormal eigenpairs

$$(\lambda_1, \mathbf{v}_1) = (16, \frac{1}{5} \begin{pmatrix} 4 \\ 3 \end{pmatrix}), \quad (\lambda_2, \mathbf{v}_2) = (-9, \frac{1}{5} \begin{pmatrix} -3 \\ 4 \end{pmatrix})$$

In the rotated basis, we have the canonical hyperbola

$$16t_1^2 - 9t_2^2 = 144 \iff \frac{t_1^2}{3^2} - \frac{t_2^2}{4^2} = 1$$

which is easily plotted.



Exercises. 6.8.1. Prove that the sum of any two bilinear forms is bilinear, and that any scalar multiple of a bilinear form is bilinear: thus the set of bilinear forms on V is a vector space.
(You can't use matrices here, since V could be infinite-dimensional!)

6.8.2. Compute the matrix of the bilinear form

$$B(\mathbf{x}, \mathbf{y}) = x_1y_1 - 2x_1y_2 + x_2y_1 - x_3y_3$$

on $\mathbb{R}^3$ with respect to the basis $\beta = \left\{ \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \right\}$.

6.8.3. Check that the function $B(f, g) = f'(0)g''(0)$ is a bilinear form on the vector space of twice-differentiable functions. Find the matrix of B with respect to $\beta = \{\cos t, \sin t, \cos 2t, \sin 2t\}$ when restricted to the subspace $\text{Span } \beta$.

6.8.4. For each matrix A with real valued entries, find a diagonal matrix D and an invertible matrix Q such that $Q^T A Q = D$.

$$(a) \begin{pmatrix} 1 & 3 \\ 3 & 2 \end{pmatrix} \quad (b) \begin{pmatrix} 3 & 1 & 2 \\ 1 & 4 & 0 \\ 2 & 0 & -1 \end{pmatrix}$$

6.8.5. If K is a quadratic form and $K(\mathbf{x}) = 2$, what is the value of $K(3\mathbf{x})$?

6.8.6. If $\mathbb{F}$ does not have characteristic 2, and $K(\mathbf{x}) = B(\mathbf{x}, \mathbf{x})$ is a symmetric quadratic form, prove that we can recover the bilinear form B via

$$B(\mathbf{x}, \mathbf{y}) = \frac{1}{2} (K(\mathbf{x} + \mathbf{y}) - K(\mathbf{x}) - K(\mathbf{y}))$$

6.8.7. If $B(\mathbf{x}, \mathbf{y}) = \mathbf{x}^T \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \mathbf{y}$ is a bilinear form on $\mathbb{F}^2$, compute the quadratic form $K(\mathbf{x})$.

6.8.8. Describe and plot the following conics:

(a) $x^2 + y^2 + xy = 6$

(b) $35x^2 + 120xy = 4x + 3y$

6.8.9. Suppose that a conic C in $\mathbb{R}^2$ has the form $ax^2 + 2bxy + cy^2 + dx + ey + f = 0$, where at least one of $a, b, c \neq 0$, and define $\Delta = b^2 - ac$. Prove that:

- C is a parabola if and only if $\Delta = 0$;
- C is an ellipse if and only if $\Delta < 0$;
- C is a hyperbola if and only if $\Delta > 0$.

(Hint: λ_1, λ_2 are the eigenvalues of a symmetric matrix, so...)

6.11 The Geometry of Orthogonal Operators

The purpose of this section is to classify the orthogonal operators $T \in \mathcal{L}(V)$ on any finite-dimensional real inner product space V . This is very easy to do when $\dim V = 1$ or 2 .

Dimension 1: Clearly $T(\mathbf{x}) = a\mathbf{x}$ for some $a \in \mathbb{R}$. T is clearly self-adjoint and therefore orthogonal if and only if $a^2 = 1$, yielding $T = \pm I$.

Dimension 2: Any choice of orthonormal basis β generates an isomorphism of inner product spaces $\phi : V \rightarrow \mathbb{R}^2$ and the problem becomes that of identifying every orthogonal matrix in $M_2(\mathbb{R})$. We already did this in Exercise 6.5.6.: all orthogonal 2×2 real matrices have the form

$$A_\theta = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}, \quad B_\theta = \begin{pmatrix} \cos \theta & \sin \theta \\ \sin \theta & -\cos \theta \end{pmatrix}$$

for some $\theta \in \mathbb{R}$. Moreover, their effects are as follows:

- $L_{A_\theta} \in \mathcal{L}(\mathbb{R}^2)$ rotates vectors counter-clockwise about the origin by θ radians;
- $L_{B_\theta} \in \mathcal{L}(\mathbb{R}^2)$ reflects vectors across the line making angle $\frac{1}{2}\theta$ with the positive x -axis.

These maps provides the motivation for a general definition of rotation and reflection.

Definition 6.82. Suppose $T \in \mathcal{L}(V)$ where V is a finite-dimensional real inner product space. We say that T is a:

Rotation if either T is the identity map, or if there exists a two-dimensional subspace $W \leq V$ with orthonormal basis $\beta = \{\mathbf{x}, \mathbf{y}\}$ and some $\theta \in (0, 2\pi)$ such that^a

$$T(\mathbf{x}) = \cos \theta \mathbf{x} + \sin \theta \mathbf{y}, \quad T(\mathbf{y}) = -\sin \theta \mathbf{x} + \cos \theta \mathbf{y}, \quad T(\mathbf{w}^\perp) = \mathbf{w}^\perp,$$

for all $\mathbf{w}^\perp \in W^\perp$, termed the *axis of rotation*. Otherwise said, $[T|_W]_\beta = A_\theta$ and $T|_{W^\perp} = I$.

Reflection if there exists a one-dimensional subspace $W \leq V$ such that $\forall \mathbf{w} \in W, \mathbf{w}^\perp \in W^\perp$,

$$T(\mathbf{w} + \mathbf{w}^\perp) = -\mathbf{w} + \mathbf{w}^\perp$$

Otherwise said, $T|_W = -I$ and $T|_{W^\perp} = I$. We could call this *reflection across $W^\perp$* .

^aWe could include the identity here with $\theta = 0$, but then any 2-dimensional subspace W is suitable and the axis of rotation is not well-defined.

Example 6.83. We compute the matrix of the reflection across $W^\perp = \text{Span}\{\mathbf{i} + 2\mathbf{j}\} \leq \mathbb{R}^2$.

First observe that $W = \text{Span}\{2\mathbf{i} - \mathbf{j}\}$, whence $\beta = \left\{ \begin{pmatrix} 2 \\ -1 \end{pmatrix}, \begin{pmatrix} 1 \\ 2 \end{pmatrix} \right\}$ is an orthogonal basis (it doesn't need to be orthonormal!). The change of co-ordinate matrix from β to the standard basis $\epsilon = \{\mathbf{i}, \mathbf{j}\}$ is $Q = Q_\beta^\epsilon = \begin{pmatrix} 2 & 1 \\ -1 & 2 \end{pmatrix}$, whence the required matrix is

$$A = Q_\beta^\epsilon \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} Q_\epsilon^\beta = \frac{1}{5} \begin{pmatrix} -3 & 4 \\ 4 & 3 \end{pmatrix}$$

which is easily checked to satisfy $A \begin{pmatrix} 2 \\ -1 \end{pmatrix} = - \begin{pmatrix} 2 \\ -1 \end{pmatrix}$ and $A \begin{pmatrix} 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$.

It should be clear that every rotation and reflection is an orthogonal operator and that they may be easily distinguished by determinant:

$$\det(\text{rotation}) = 1, \quad \det(\text{reflection}) = -1$$

It is also easy to see that $L_{A_\theta} \in \mathcal{L}(\mathbb{R}^2)$ is a rotation as per the definition: simply take $\beta = \{\mathbf{i}, \mathbf{j}\}$ and $W = \mathbb{R}^2$! We leave the verification that $L_{B_\theta} \in \mathcal{L}(\mathbb{R}^2)$ is a reflection as an exercise. The case for $\dim V = 1$ is even easier. To summarize:

Lemma 6.84. *Suppose V is a real inner product space with $\dim V = 1$ or 2 . Then every orthogonal operator $T \in \mathcal{L}(V)$ is either a rotation or a reflection.*

Our goal is to describe any orthogonal operator T in terms of rotations and reflections. The upshot is that there exists an orthonormal basis β with respect to which the matrix of T has block form

$$[T]_\beta = \text{diag}(R_1, R_2, \dots, R_k)$$

and each R_j is either a 2×2 rotation matrix, or is ± 1 .

Example 6.85. The matrix $A = \begin{pmatrix} \frac{3}{5} & -\frac{4}{5} & 0 & 0 \\ \frac{4}{5} & \frac{3}{5} & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}$ describes an orthogonal operator $L_A \in \mathcal{L}(\mathbb{R}^4)$. We have three commuting orthogonal operations $L_A = T_1 T_2 T_3$:

- T_1 rotates by $\theta = \tan^{-1} \frac{4}{3}$ about the axis $W_1^\perp = \text{Span}\{\mathbf{e}_3, \mathbf{e}_4\}$;
- T_2 is the identity rotation with ‘axis’ $W_2^\perp = \text{Span}\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_4\}$;
- T_3 is reflection across $W_3^\perp = \text{Span}\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$

To see that this always happens, we first need a little theory regarding a real $n \times n$ matrix A . Our primary strategy always relies on eigendecompositions; unfortunately A need not have any *real* eigenvectors. As a linear operator on $\mathbb{C}^n$ however, it must have at least one eigenpair $(\lambda, \mathbf{v})$: consider the real and imaginary parts: $\mathbf{v} = \mathbf{x} + i\mathbf{y}$ and $\lambda = p + iq$ and observe that

$$A\mathbf{x} + iA\mathbf{y} = A(\mathbf{x} + i\mathbf{y}) = (p + iq)(\mathbf{x} + i\mathbf{y}) = (p\mathbf{x} - q\mathbf{y}) + i(p\mathbf{y} + q\mathbf{x}) \quad (*)$$

Since A is real, we can equate real and imaginary parts to see that $\text{Span}\{\mathbf{x}, \mathbf{y}\}$ is an L_A -invariant subspace of $\mathbb{R}^n$. This is the generic situation, since, upon choosing a basis, any linear operator on a finite-dimensional real vector space is multiplication by some real matrix A . We therefore conclude:

Lemma 6.86. *If $T \in \mathcal{L}(V)$ is a linear operator on a finite-dimensional real vector space, then there exists a T -invariant subspace $W \leq V$ such that $\dim W = 1$ or 2 .*

The reason our construction might produce a 1-dimensional space? The real and imaginary parts $\mathbf{x}, \mathbf{y}$ might be parallel! In this case, of course, W is an eigenspace of T .

Of particular interest to our current goal is that the real and imaginary parts of $(*)$ look very like a *rotation*. Indeed if we specialize the Lemma to an orthogonal operator T , it is an easy exercise to check

that $W^\perp$ is also T -invariant. We conclude that $T|_{W^\perp}$ is an orthogonal operator on a lower-dimensional vector space. We may now repeat the process until we have exhausted V . This leads to the following:

Theorem 6.87. *Let $T \in \mathcal{L}(V)$ be an orthogonal operator on a finite-dimensional real inner product space. Then:*

1. *There exists an orthogonal decomposition*

$$V = W_1 \oplus \cdots \oplus W_m$$

where each W_k is a 1 or 2-dimensional T -invariant subspace.^a

2. *For each j , define $T_j := T|_{W_j} + I|_{W_j^\perp}$. Then T_j is a rotation or a reflection and $T = T_1 \cdots T_m$.*
3. *$\det T = (-1)^k$ where exactly k of the T_j are reflections.*
4. *The subspaces may be chosen so that there is at most one reflection T_j , and where $\dim W_j = 1$.*

^aIt should be clear that $\frac{n}{2} \leq m \leq n$.

Proof. 1. We've already done this, though it could be made more formal by induction.

2. Since $T|_{W_j}$ is orthogonal on a 1 or 2-dimensional space, Lemma 6.84 says it is either a rotation or a reflection. Extending to T_j by forcing the identity on $W_j^\perp$ produces a rotation or reflection on $\mathbb{R}^n$. Moreover, since the W_j are orthogonal subspaces, we see that

$$\mathbf{v}_j \in W_j \implies T_k(\mathbf{v}_j) = \begin{cases} T(\mathbf{v}_j) & \text{if } j = k \\ \mathbf{v}_j & \text{if } j \neq k \end{cases}$$

By applying to a basis where each $\mathbf{v}_j$ lies in some W_j , the relationship $T = T_1 \cdots T_m$ is now clear.

3. This is immediate since every rotation has determinant 1 and every reflection determinant -1 .
4. Suppose that T_j is a reflection and $\dim W_j = 2$. Then there exists a one-dimensional subspace $\hat{W}_j$ such that

$$T_j|_{\hat{W}_j} = -I, \quad T_j|_{\hat{W}_j^\perp} = I$$

By replacing W_j by $\hat{W}_j \oplus \hat{W}_j^\perp$, we see that every reflection may be assumed to be one-dimensional.

Now suppose W_j, W_k are 1-dimensional subspaces corresponding to reflections T_j, T_k . Then

$$(T_j + T_k)(\mathbf{v}) = \begin{cases} -\mathbf{v} & \text{if } \mathbf{v} \in W_j \oplus W_k \\ \mathbf{v} & \text{if } \mathbf{v} \in (W_j \oplus W_k)^\perp \end{cases}$$

Otherwise said, $T_j + T_k$ is the *rotation* with $W = W_j \oplus W_k$ and $\theta = \pi$. We can therefore replace an even number of reflections with rotations. ■

Example 6.88. Consider the matrix $A = \frac{1}{25} \begin{pmatrix} 15 & -12 & -16 \\ 20 & 9 & 12 \\ 0 & -20 & 15 \end{pmatrix}$. We show that $L_A \in \mathcal{L}(\mathbb{R}^3)$ is a rotation and find its axis and angle of rotation.

It is easy (with a calculator!) to check that $A^T A = I$ so that A is orthogonal, and moreover that $\det A = 1$. According to the Theorem, there are no reflections in A and so there must exist a decomposition $\mathbb{R}^3 = W \oplus W^\perp$ where $\dim W = 2$ and such that $W^\perp$ is an eigenspace with eigenvalue 1. We can hunt for the eigenvalues via the usual approach, or we can do it by brute force: $\mathbf{w}^\perp = \begin{pmatrix} a \\ b \\ c \end{pmatrix}$ has eigenvalue 1 if and only if

$$\begin{cases} 15a - 12b - 16c = 25a \\ 20a + 9b + 12c = 25b \\ -20b + 15c = 25c \end{cases} \iff c = -2b, a = 2b \iff \mathbf{w}^\perp \propto \begin{pmatrix} 2 \\ 1 \\ -2 \end{pmatrix}$$

The axis of rotation is therefore $W^\perp = \text{Span} \left\{ \begin{pmatrix} 2 \\ 1 \\ -2 \end{pmatrix} \right\}$. To find the angle, we simply need to choose anything orthogonal to this (and therefore in W), and see how A changes the angle. Thus let $\mathbf{w} = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$ and compute

$$A\mathbf{w} = \frac{1}{25} \begin{pmatrix} -1 \\ 32 \\ 15 \end{pmatrix} \implies \cos \theta = \frac{\begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} -1 \\ 32 \\ 15 \end{pmatrix}}{\sqrt{1+1}\sqrt{1+32^2+15^2}} = \frac{14}{50} = \frac{7}{25} \implies \theta \approx 74^\circ$$

In fact $(\mathbf{w}, A\mathbf{w}, \mathbf{w}^\perp)$ is negatively oriented if we ‘look down’ on $W = \text{Span}\{\mathbf{w}, A\mathbf{w}\}$ from above $\mathbf{w}^\perp$, the rotation would be through 74° *clockwise*.

Because of the theorem, some authors prefer to simply define a rotation to be any orthogonal operator with determinant 1 and a reflection to be any with determinant -1 . The multiplicative property of determinant then shows that the composition of any two rotations is a rotation,⁵ of two reflections is a rotation, etc. The problem with this definition is that one loses the notion of an axis of rotation, since a general determinant 1 orthogonal operator will typically require multiple rotations around several axes.

Example 6.89. For given θ, ϕ , the matrix

$$A = \begin{pmatrix} \cos \theta & -\sin \theta & 0 & 0 \\ \sin \theta & \cos \theta & 0 & 0 \\ 0 & 0 & \cos \phi & -\sin \phi \\ 0 & 0 & \sin \phi & \cos \phi \end{pmatrix}$$

performs two rotations. It has no ‘axis’ of rotation, since it has no eigenspace with eigenvalue 1: indeed it has no *real* eigenvalues at all. Regardless, it is orthogonal and has determinant 1, so many authors consider this to be a rotation.

⁵In particular, this shows that the set of ‘rotations’ forms a normal subgroup $\text{SO}(V)$ of the group of orthogonal operators on V , known as the *special orthogonal group*.

Exercises. 6.11.1. Find the matrices with respect to the standard basis of the reflections in $W^\perp$:

(a) $W = \text{Span}\{-2\mathbf{i} + 5\mathbf{j}\}$

(b) $W = \text{Span}\{\mathbf{i} + 2\mathbf{j} + 2\mathbf{k}\}$ *Hint: consider the basis* $\beta = \left\{ \frac{1}{3} \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix}, \frac{1}{3} \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix}, \frac{1}{3} \begin{pmatrix} 2 \\ 1 \\ -2 \end{pmatrix} \right\}$

6.11.2. (a) Given ϕ , let $\beta = \{\mathbf{x}, \mathbf{y}\} = \left\{ \begin{pmatrix} \cos \phi \\ \sin \phi \end{pmatrix}, \begin{pmatrix} -\sin \phi \\ \cos \phi \end{pmatrix} \right\}$ and $\gamma = \{\mathbf{x}, -\mathbf{y}\}$. Clearly every orthonormal basis of $\mathbb{R}^2$ has the form β or γ . Compute the matrices

$$[L_{A_\theta}]_\beta \quad \text{and} \quad [L_{A_\theta}]_\gamma \quad \text{where} \quad A_\theta = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$$

What do you observe?

(b) Now consider the reflection matrix $B_\theta = \begin{pmatrix} \cos \theta & \sin \theta \\ \sin \theta & -\cos \theta \end{pmatrix}$. Show that there is a privileged orthonormal basis (indeed four up to scaling and reordering) β such that $[L_{B_\theta}]_\beta = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$.

6.11.3. Compute $A_\phi B_\theta A_\phi^{-1}$. Across what line in $\mathbb{R}^2$ does this matrix reflect? Can you do this *without* multiplying out the matrices?

6.11.4. Prove explicitly:

(a) If T is a rotation then it is orthogonal and $\det T = 1$;

(b) If T is a reflection then it is orthogonal and $\det T = -1$.

6.11.5. If W is a T -invariant subspace where T is an orthogonal operator on a finite-dimensional real inner product space V , prove that $W^\perp$ is also T -invariant.

6.11.6. Find the result when the vector $\mathbf{x} = \begin{pmatrix} 3 \\ 4 \\ 2 \end{pmatrix}$ is rotated 90° counter-clockwise around the z -axis (when viewed from above) and then reflected in the xz -plane. What is the effect of this transformation on a general vector?

6.11.7. For any non-zero $\theta, \phi \in (-\pi, \pi)$, define the matrices

$$A = \begin{pmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad B = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos \phi & -\sin \phi \\ 0 & \sin \phi & \cos \phi \end{pmatrix}$$

Prove that L_{AB} is a rotation of $\mathbb{R}^3$ and find its axis of rotation.

6.11.8. Let V be a real inner product space of dimension 2. Given any $\mathbf{x} \neq \mathbf{y}$ with $\|\mathbf{x}\| = \|\mathbf{y}\| = 1$, show that there exists a unique rotation $T \in \mathcal{L}(V)$ such that $T(\mathbf{x}) = \mathbf{y}$.